# Unit - IV

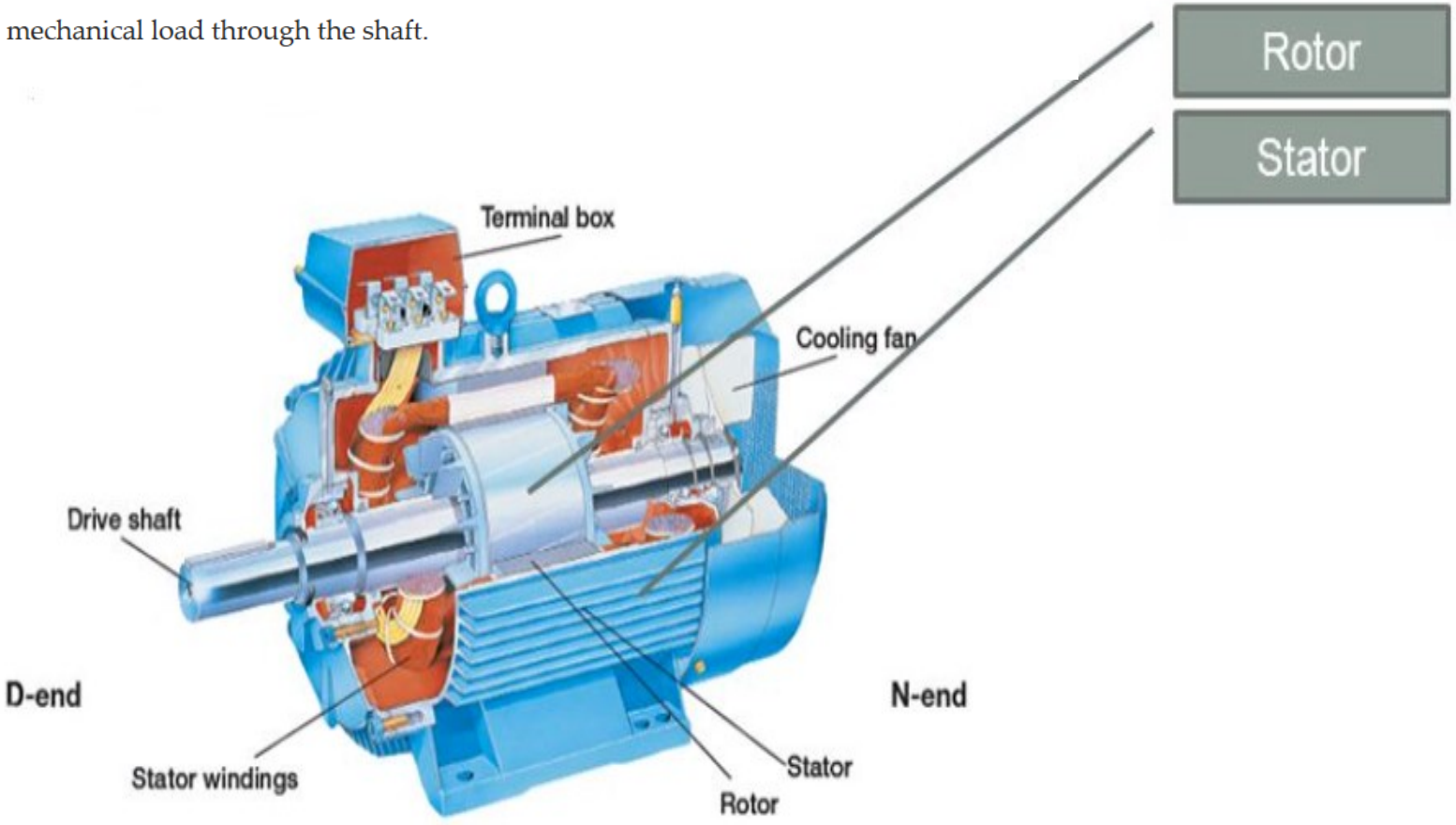
# 3- ϕ Induction Motor

- An **electric motor** is a device which converts an electrical energy into a mechanical energy. This mechanical energy then can be supplied to various types of loads. The motors can operate **on DC** as well as **1-ϕ and 3- ϕ** supply.
- The motors operating on DC supply are called **DC motors**, while motors operating on AC supply are called **AC motors**.
- The AC motors are classified as **1-ϕ** and **3-ϕ *Induction motors***, *Synchronous motors* and some special purpose motors. Out of all these types, **3-ϕ *Induction motors*** are widely used for various industrial applications.
- The important ***Advantages*** of **3-ϕ *Induction motors*** over other types are **Self starting property**, ***N*o need of starting device**, ***H*igh power factor**, ***G*ood speed regulation** and ***R*obust construction**.

# 3- ϕ Induction Motor Construction

Like any other type of electrical motor induction motor, a 3 phase induction motor is constructed from two main parts, namely the rotor and stator:

1. Stator: As its name indicates stator is a stationary part of induction motor. A stator winding is placed in the stator of induction motor and the three phase supply is given to it.
2. Rotor: The rotor is a rotating part of induction motor. The rotor is connected to the mechanical load through the shaft.



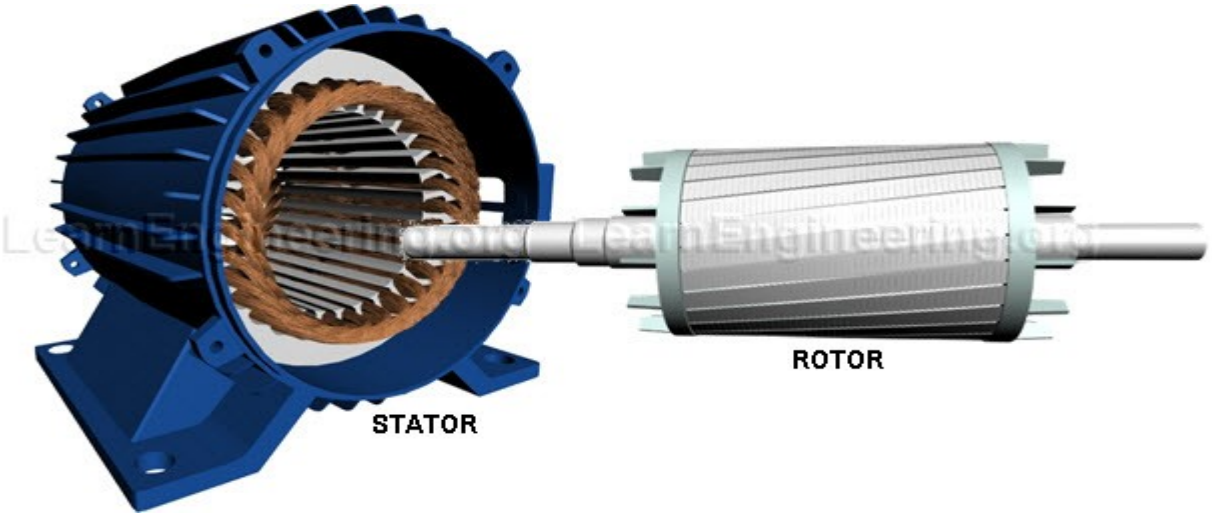
## Stator:

- Stator of an IM consists of steel frame which contains hallow cylindrical hole made up of thin laminations of silicon steel to reduce hysteresis and eddy current losses.
- The number of evenly spaced slots are provided on ***inner*** periphery of the laminations.
- The insulated conductors are placed in the stator slots and are suitably connected to form a balanced **3- ϕ star or delta** connections.

## Rotor:

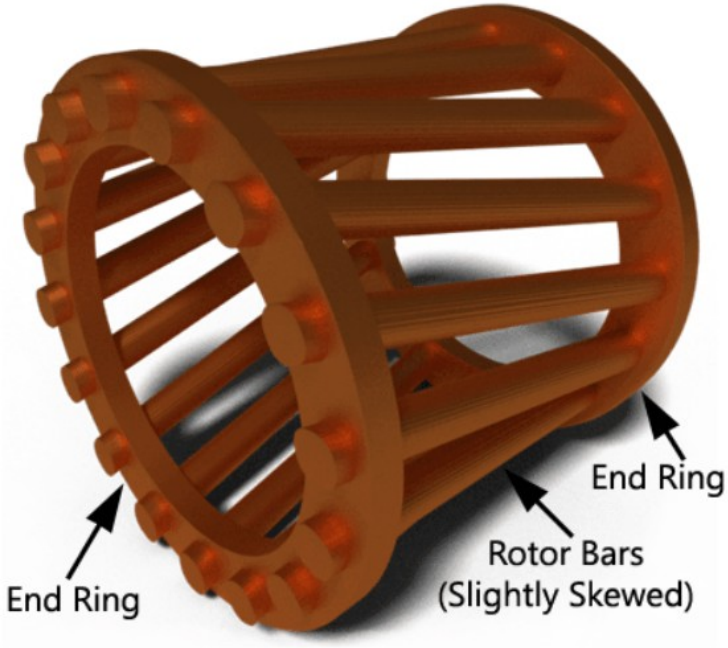
- **Rotor is placed inside the stator**.
- Rotor is mounted on the shaft of a hallow laminated core having slots on its ***outer*** periphery, the windings are placed in these slots are called rotor windings and they may be of many types like:

- ***Squirrel Cage Rotor Type IM***
- ***Slip Ring Rotor Type IM***

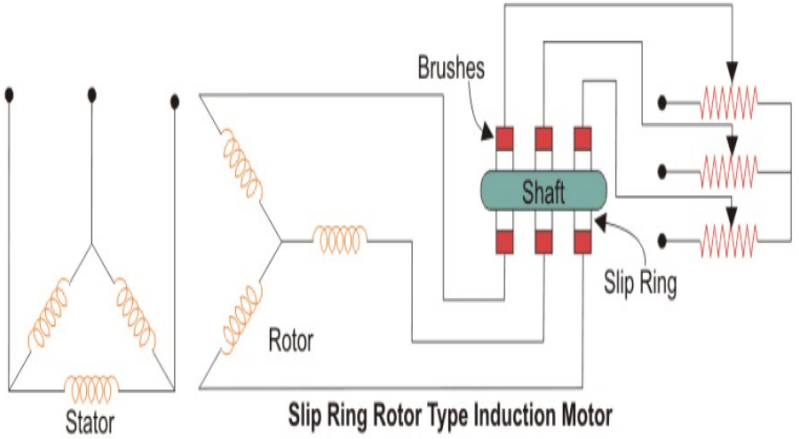


# Types of 3- ϕ Induction Motor

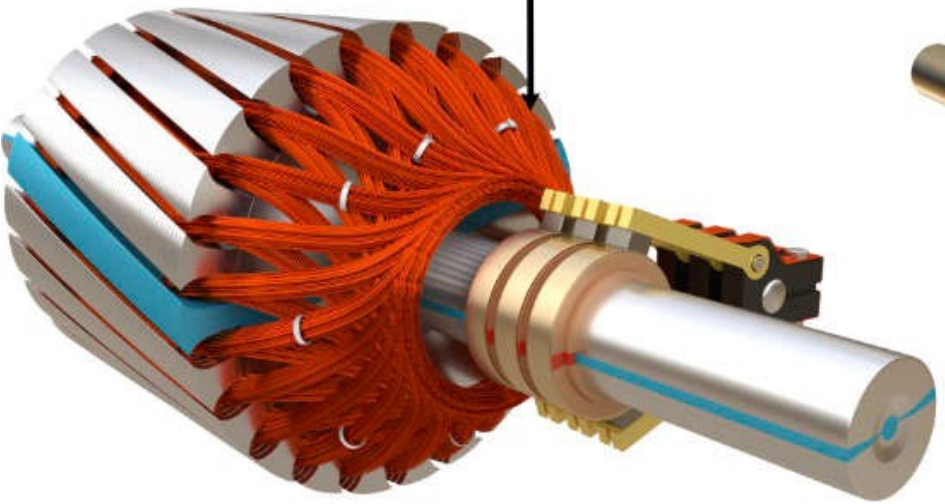
## Squirrel Cage Induction Motor




## Slip Ring Induction Motor

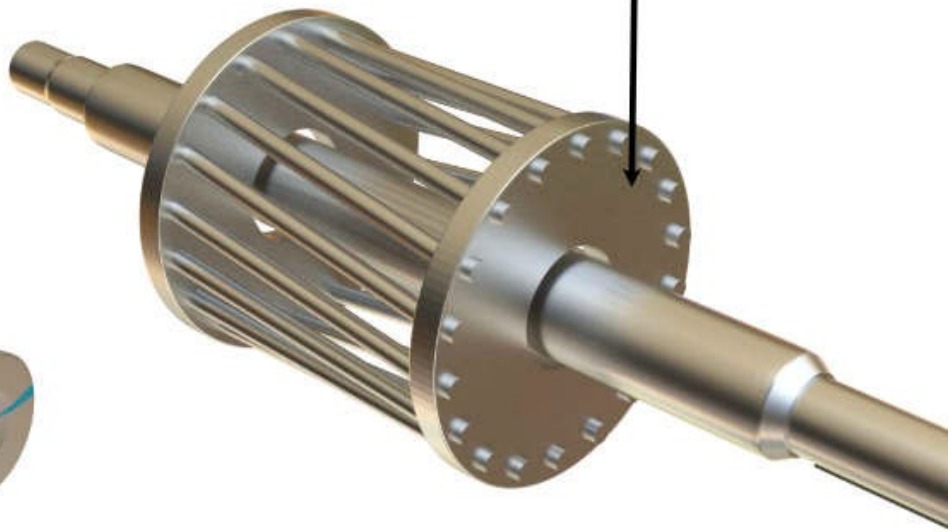


HIGH STARTING TORQUE



SLIP RING ROTOR

LOW STARTING TORQUE



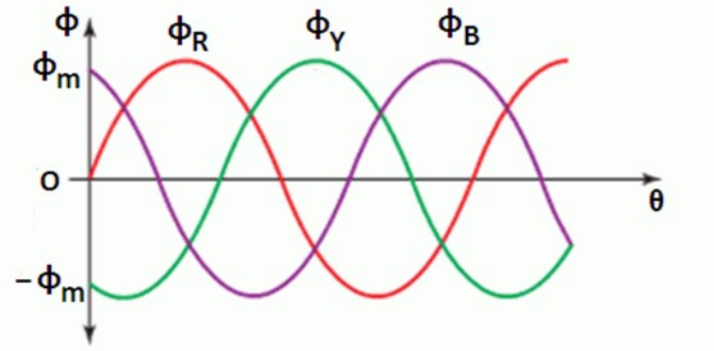
SQUIRREL CAGE ROTOR

# Types of 3- ϕ Induction Motor

| Squirrel Cage Rotor | Slip Ring Rotor |
|---|---|
| Rotor consists of bars which are shorted at the ends with the help of end rings | Rotor consists of a three phase winding similar to the stator winding |
| Construction is very simple. As permanently shorted, external resistance cannot be added | Construction is complicated. Resistance can be added externally |
| **Very common and almost 95% induction motors use this type of rotor** | **Only 5% of induction motors in industry use slip ring rotor** |
| Rotor copper **losses are less**, hence have **higher efficiency.** | Rotor copper **losses are high** , hence **efficiency is less** |
| Used for Lathes, drilling machines, Fans, Blowers, Water pumps, Grinders and Printing machines etc | Used for Lifts, Hoists, Cranes, Elevators and Compressors etc |

# Principle of Operation of 3- ϕ Induction Motor

- ***Induction motor*** works on the principle of ***Electromagnetic induction***.
- The working principle of ***3- ϕ Induction motors*** is based on the production of ***Rotating Magnetic Field*** (***RMF***).
- When a 3-ϕ supply is given to the 3- ϕ set of stator winding, a rotating magnetic field of constant magnitude is produced. The speed of this rotating magnetic field is synchronous speed, **Ns = 120f/P rpm.**
- The current carrying windings produce the magnetic field or flux. And *due to interaction of three fluxes produced due to 3- ϕ supply,* ***resultant flux has a constant magnitude (1.5$\phi_m$) and its axis rotating in space, without physically rotating three windings.***
- This type of field is nothing but ***Rotating Magnetic Field (RMF).***



Fluxes produced by line currents

# Principle of Operation of 3- ϕ Induction Motor (Cont…)

- Now at this instant rotor is stationary and stator flux RMF is rotating. So its obvious that there exists a relative motion between the RMF and rotor conductors.
- Now RMF gets cut by the rotor conductors as RMF sweeps over rotor conductors. **Whenever conductor cuts the flux, EMF gets induced in it called Rotor Induced EMF (Faradays laws).**
- As **rotor forms closed circuit**, induced emf circulates current through rotor called rotor current.
- So rotor conductors are placed in magnetic field and experiences a force to rotate.
- As all the rotor conductors experiences a force, the overall rotor experiences a torque and starts rotating, So interaction of the two fluxes is very essential for a motoring action.
- Hence ***Rotor starts rotating in the same direction as that of Rotating Magnetic Field.***

- For a standard frequency whatever **speed of *RMF*** results is called Synchronous Speed, in case of induction motors, it is denoted as **'Ns'**
- For a **4 pole** machine, 415V, 50Hz, what is my **Ns?**
  - **Ns = (120f/P) = Speed of RMF = (120X50)/4= 1500**

f- supply frequency

P= number of poles for which stator winding is wounded

***This is the speed with which RMF rotates in space.***

## Applications of Induction Motors:

- Fan, Blowers, Machine tools
- Compressor, Crushers, Conveyors,
- Cranes, Compressors, Elevators, Hoist etc

# Slip of 3- ϕ Induction Motor

- So $N_s$ = Speed of rotating magnetic field (RMF) in rpm
- **N** = speed of rotor i.e motor in rpm

(**Ns-N)** = **Relative speed between** the two speed is called **Slip speed**

- In practical, rotor continues to rotate with a speed **slightly less** than the synchronous speed of the rotating magnetic field (**N < Ns**) .
- Induction motor never rotates at synchronous speed.

- **S = (Ns-N)/Ns** ---------- absolute slip
- **%S = (Ns-N)/(Ns)*100**---------- percentage slip

- ***At starting, actual speed N is zero , so S=1***

# Rotor Frequency

In case of induction motor, the speed of rotating magnetic field is,

$N_s = (120\,f)/P$ .....(1)

Where f = Frequency of supply in Hz

At start when N = 0, s = 1 and stationary rotor has maximum relative motion with respect to R.M.F. Hence maximum e.m.f. gets induced in the rotor at start. The frequency of this induced e.m.f. at start is same as that of supply frequency.

As motor actually rotates with speed N, the relative speed of rotor with respect R.M.F. decreases and becomes equal to slip speed of $N_s - N$. The induced e.m.f. in rotor depends on rate of cutting flux i.e. relative speed $N_s - N$.

Hence in running condition magnitude of induced e.m.f. decreases so as to its frequency. The rotor is wound for same number of poles as that of stator i.e. P. If $f_r$ is the frequency of rotor induced e.m.f. in running condition at slip speed $N_s - N$ then there exists a fixed relation between $(N_s - N)$, $f_r$ and P similar to equation (1). So we can write for rotor in running condition,

$(N_s - N) = (120\,f_r)/P$ , rotor poles = stator poles = P .....(2)

Dividing (2) by (1) we get,

$(N_s - N)/N_s = (120\,f_r / P)/(120\,f / P)$ but $(N_s - N)/N_s$ = slip s

$s = f_r/f$

$f_r = s\,f$

# Numerical problems

- A 4 pole, 3 phase induction motor is supplied from 50 Hz supply. Determine its synchronous speed. On full load, its speed is observed to be 1410 rpm. Calculate its full load slip.

**P= 4, f= 50 Hz, N=1410 rpm, S=?**

Ns = (120Xf)/P = 1500 rpm

**S = (Ns-N)/Ns = 0.06**

**%S = (0.06X100) = 6%**

- A 4 pole, 3 phase, 50 Hz, star connected induction motor has a full load slip of 4%. Calculate full load speed of the motor.
- **%S = 4, P=4, f=50Hz, S= 0.04**
- **N = ?**
- Ns = (120Xf)/P = 1500 rpm
- **S = (Ns-N)/Ns**

**0.04 = (1500-N)/1500**

**N = 1440 rpm**

**Example**: A 4 pole, 3 phase, 50 Hz induction motor runs at a speed of 1470 r.p.m. speed. Find the frequency of the induced e.m.f in the rotor under this condition.

**Solution** : The given values are,

P = 4, f = 50 Hz, N = 1470 r.p.m.

$N_s$ = (120 f )/ P = (120 x 50)/4 = 1500 r.p.m.

s = ($N_s$ – N)/$N_s$ = (1500-1470)/1500 = 0.02

$f_r$ = s f = 0.02 x 50 = 1 Hz

It can be seen that in running condition, frequency of rotor induced e.m.f. is very small.

# Numerical Problems

**Example (3)**

*A 4-pole, 3-phase induction motor operates from a supply whose frequency is 50 Hz. Calculate :* ***(i)*** *the speed at which the magnetic field of the stator is rotating.* ***(ii)*** *the speed of the rotor when the slip is 0.04.* ***(iii)*** *the frequency of the rotor currents when the slip is 0.03.* ***(iv)*** *The frequency of the rotor currents at standstill.*

***(i)*** Stator field revolves at synchronous speed, given by

$Ns = 60\, f_1/P = 60 \times 50/2 =$ **1500 r.p.m.**

***(ii)*** rotor (or motor) speed, $N = Ns\,(1 - s) = 1500(1 - 0.04) =$ **1440 r.p.m.**

***(iii)*** frequency of rotor current, $f_2 = sf_1 = 0.03 \times 50 = 1.5$ c.p.s = **1.5 Hz**

***(iv)*** Since at standstill, $s = 1, f_2 = sf_1 = 1 \times f_1 =$ **50Hz**

**Example (4)**

*A 3-φ induction motor is wound for 4 poles and is supplied from 50-Hz system. Calculate* ***(i)*** *the synchronous speed* ***(ii)*** *the rotor speed, when slip is 4% and* ***(iii)*** *rotor frequency*

*when rotor runs at 600 rpm.*

***(i)*** $Ns = 60\, f_1/P = 60 \times 50/2 = 1500$ rpm

***(ii)*** rotor speed, $N = Ns\,(1 - s) = 1500\,(1 - 0.04) =$ **1440 rpm**

***(iii)*** when rotor speed is 600 rpm, slip is $s = (Ns - N)/Ns = (1500 - 600)/1500 = 0.6$

rotor current frequency, $f_2 = sf_1 = 0.6 \times 50 =$ **30 Hz**

# Torque Equation of 3- ϕ Induction Motor

- The torque produced in the induction motor depends on the following factor:
  - The part of rotating magnetic field which reacts with rotor and is responsible to produce induced emf in the rotor (**ϕ)**
  - The magnitude of rotor current in running condition (**$I_2$**)
  - The power factor of the rotor circuit in running condition (**cos $\phi_2$)**

$$T \alpha \phi I_2 \cos \phi_{2r}$$

  - The flux produced by stator is proportional to stator voltage $E_1$ and rotor voltage too **$E_2$**

$$\Phi \alpha E_2$$

# Torque Slip Characteristics of Three phase IM

- As the IM is loaded from ***No load to Full load***, its speed decreases hence slip increases.

- The curve obtained by plotting torque against slip from **s =1 (start)** to **s = 0 (at synchronous speed)** is called ***Torque – Slip characteristics*.**

- At constant supply voltage, $E_2$ is also constant, so we write torque equation as

$$T \propto E_2 \frac{sE_2}{\sqrt{R_2^2+(sX_2)^2}} \times \frac{R_2}{\sqrt{R_2^2+(sX_2)^2}}$$

$$T = s.R_2 / [R_2^2+(sX_2)^2]$$

- Now to judge the nature of **torque-slip characteristics**, let us divide the slip range ( **s = 0 to s = 1**) into two parts and analyze them independently.
  (i) Low Slip Region
  (ii) High Slip Region

# Torque Slip Characteristics of Three phase IM

**(i) Low Slip Region :**

In low slip region, s is very very small. Due to this the term $(sX_2)^2$ is so small as compared to / $R_2^2$ , that it can be neglected

$$\mathbf{T \alpha\ s.}R_2/ R_2^2 \ \mathbf{\alpha\ s} \qquad \text{--- as } R_2 \text{ is constant}$$

- Hence in low slip region **torque is directly proportional to slip**.
- At $N=N_s$, $s = 0$ hence $T= 0$. As no torque is generated at N=Ns, motor stops if it tries to achieve the synchronous speed. Torque increases linearly in this region, of low slip values.
- So range **$s = 0$ to $s = s_m$** is called **low slip region**, known as **Stable region of operation.**

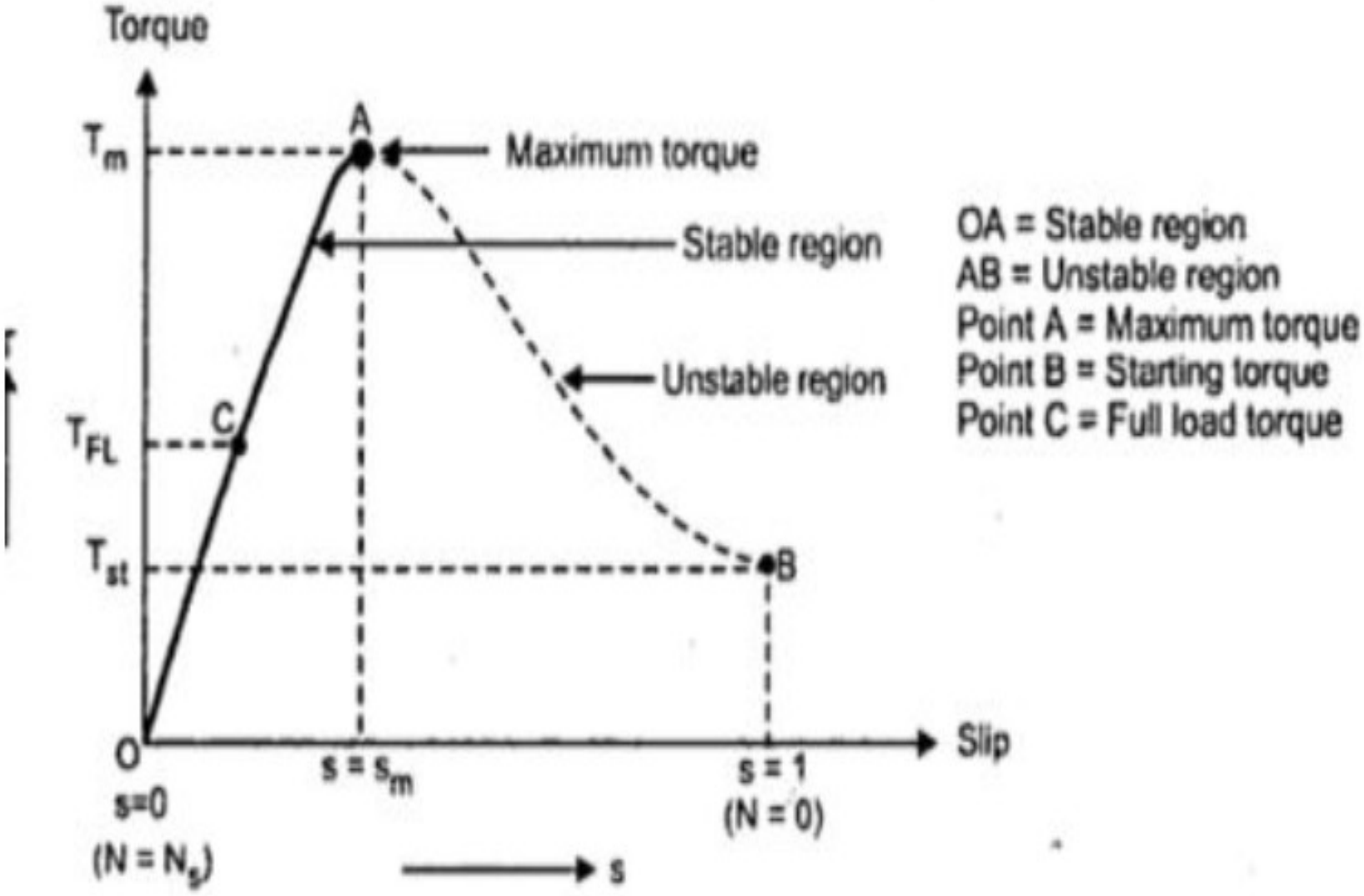
# Torque Slip Characteristics of Three Phase IM

**(ii) High Slip Region :**

- In this region, slip is very high i.e slip value is approaching to 1. Here it can be assumed that the term $\mathbf{R_2^2}$ is very very small as compared to $\mathbf{(sX_2)^2}$ Hence neglecting $\mathbf{R_2^2}$ from the denominator, we get·

$$\mathbf{T \alpha}\ s.R_2/(sX_2)^2\ \mathbf{\alpha\ 1/\ s} \qquad \text{--- as } X_2 \text{ \& } R_2 \text{ are constants}$$

- So in high slip region torque is inversely proportional to the slip. Hence it is like **rectangular hyperbola.**
- Now when load increases, load demand increases but speed decreases. As speed decrease, Slip increases, In high slip region as **T α 1/ s ,** torque decreases as slip increases. But torque must increase to satisfy the load demand. Again torque decreases as **T α 1/ s** hence same load acts as an extra load due to reduction in torque produced. Hence speed further drops. Eventually motor comes to standstill condition. The motor cannot continue to rotate at any point in this high slip region. Hence this region is called **Unstable region of operation.**
- So range **$s = s_m$ to s = 1** is called **High slip region**, known as **Unstable region of operation.**


Torque
$T_m$
A
Maximum torque
Stable region
Unstable region
$T_{FL}$
C
$T_{st}$
B
O
s=0
(N = $N_s$)
s = $s_m$
s = 1
(N = 0)
Slip
s
OA = Stable region
AB = Unstable region
Point A = Maximum torque
Point B = Starting torque
Point C = Full load torque

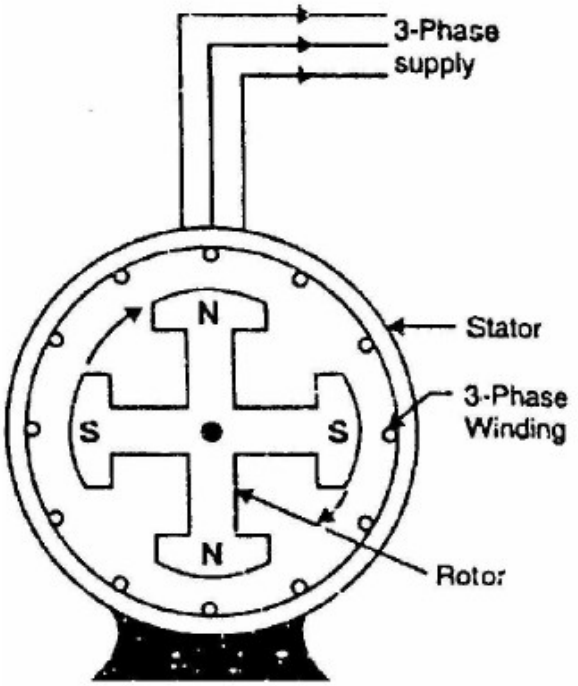
# Synchronous Generator

# (or)

# Alternator

# ( or)

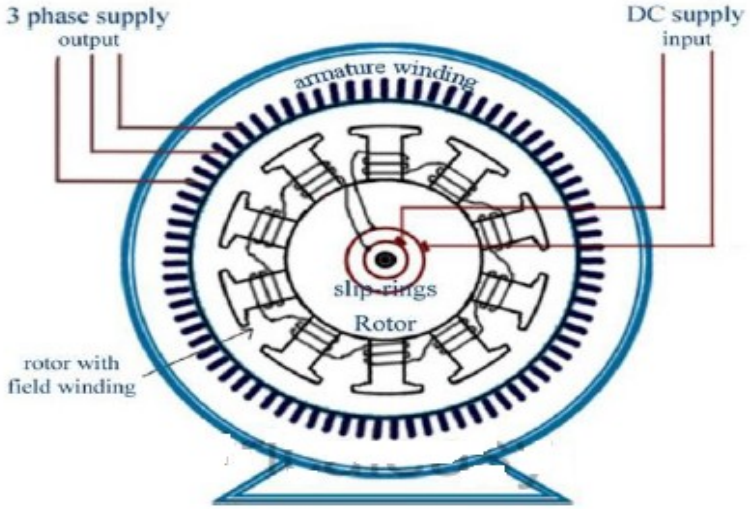
# AC Generator




A.C. system has a number of advantages over d.c. system. These days 3-phase a.c. system is being exclusively used for generation, transmission and distribution of power. The machine which produces 3-phase power from mechanical power is called an alternator or synchronous generator. Alternators are the primary source of all the electrical energy we consume. These machines are the largest energy converters found in the world. They convert mechanical energy into a.c. energy.

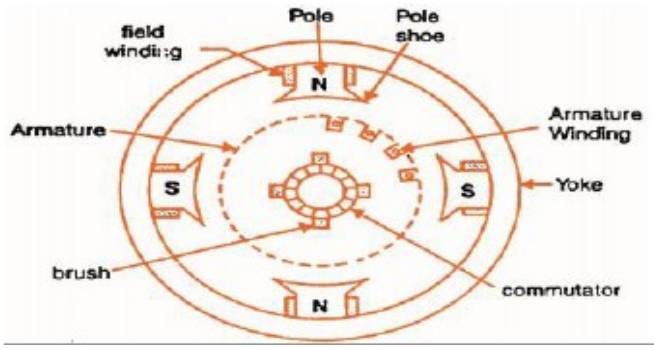
## ALTERNATOR : CONSTRUCTION

- STATOR (ARMATURE)
- ROTOR (FIELD SYSTEM)

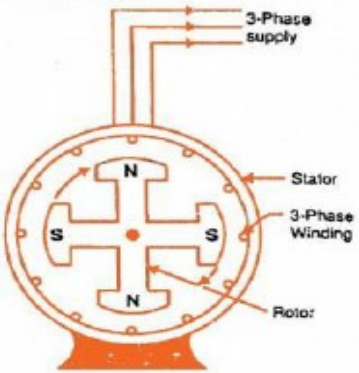



## CONSTRUCTION

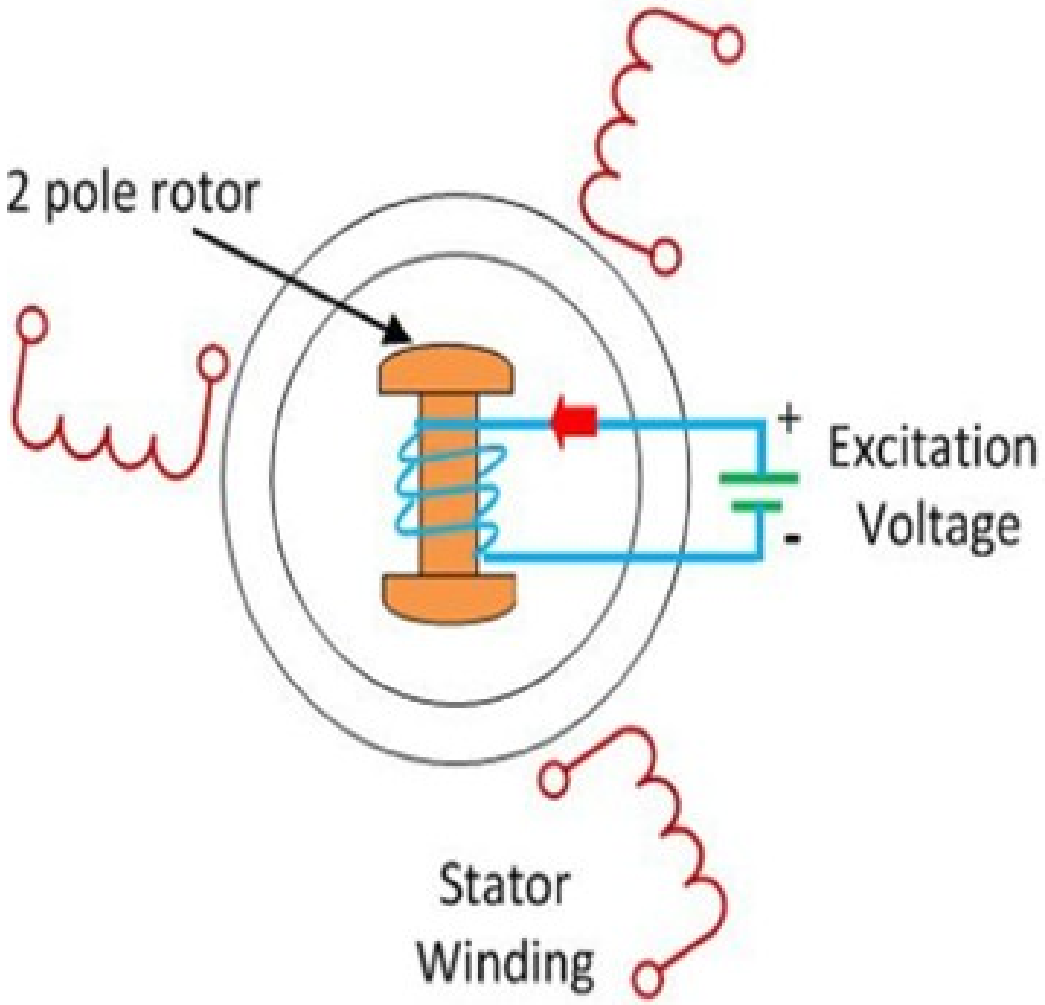
STATIONARY FIELD AND ROTATING ARMATURE TYPE: Used in DC GENERATORS



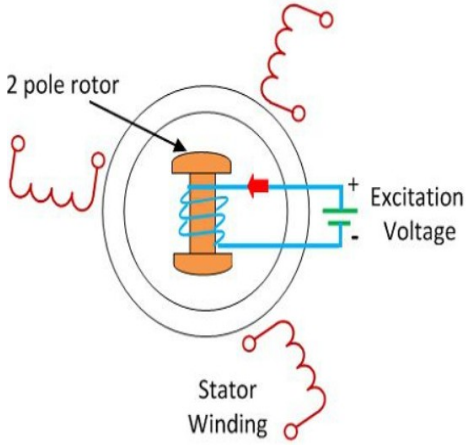

ROTATING FIELD AND STATIONARY ARMATURE TYPE: Used in ALTERNATORS



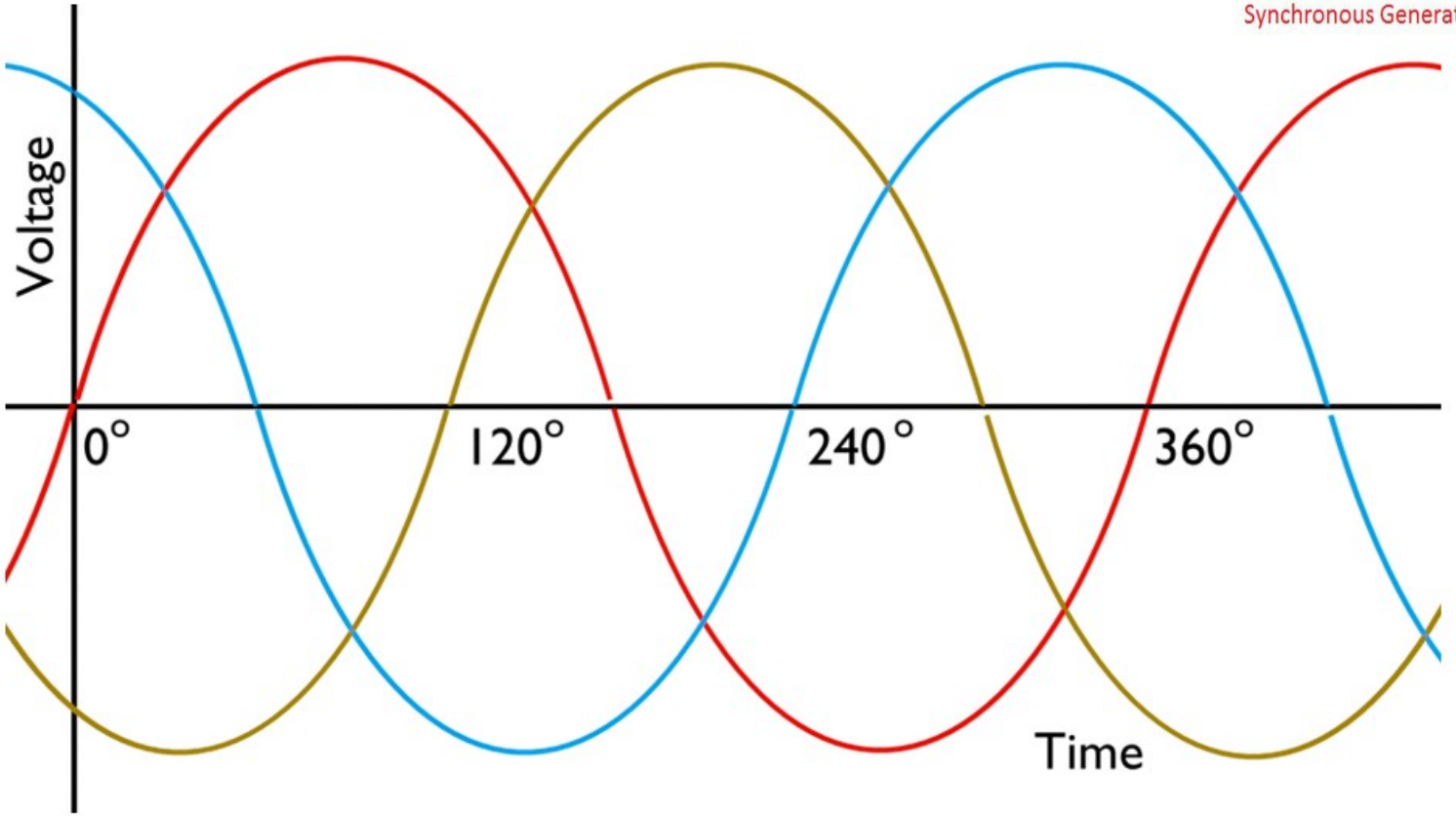
# Synchronous Generator (or) Alternator




Synchronous Generator




Synchronous Generator



# Construction of Alternator

An alternator has 3,-phase winding on the stator and a d.c. field winding on the rotor.

## 1. Stator

It is the stationary part of the machine and is built up of sheet-steel laminations having slots on its inner periphery. A 3-phase winding is placed in these slots and serves as the armature winding of the alternator. The armature winding is always connected in star and the neutral is connected to ground.

## 2. Rotor

The rotor carries a field winding which is supplied with direct current through two slip rings by a separate d.c. source. This d.c. source (called exciter) is generally a small d.c. shunt or compound generator mounted on the shaft of the alternator. Rotor construction is of two types, namely;

(i) Salient (or projecting) pole type
(ii) Non-salient (or cylindrical) pole type

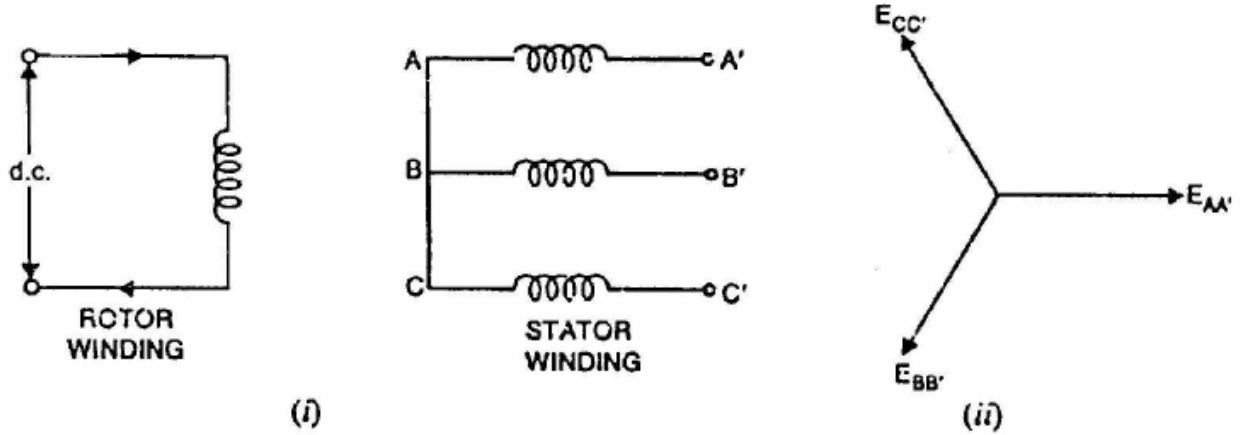
# Alternator Operation

The rotor winding is energized from the d.c. exciter and alternate N and S poles are developed on the rotor. When the rotor is rotated in anti-clockwise direction by a prime mover, the stator or armature conductors are cut by the magnetic flux of rotor poles. Consequently, e.m.f. is induced in the armature conductors due to electromagnetic induction. The induced e.m.f. is alternating since N and S poles of rotor alternately pass the armature conductors. The direction of induced e.m.f. can be found by Fleming's right hand rule and frequency is given by;

$$f = \frac{NP}{120}$$

where N = speed of rotor in r.p.m.
P = number of rotor poles

The magnitude of the voltage induced in each phase depends upon the rotor flux, the number and position of the conductors in the phase and the speed of the rotor.



# Principle of Operation of Alternator

- The **working principle of alternator** is very simple. It is just like basic principle of **DC Generator**. It also depends upon Faradays laws of electromagnetic induction which says the current is induced in the conductor inside a magnetic field when there is a relative motion between that conductor and the magnetic field. As armature is connected to commutator in a d.c generator, armature must be rotating member while field as a stationary. But ***in case of Alternators*** it is possible to have

***Rotating field & Stationary armature***

- ***The generation level of AC voltage may be higher as 11 kV to 33 kV.*** This gets induced in the armature. For stationary armature large space can be provided to accommodate large number of conductors and the insulation.
- ***It is easier to collect large currents at very high voltages from a stationary member than from slip ring and brush assembly***.
- Due to low voltage level of the field side, the insulation required is less and hence field system has very low inertia.
- If field is rotating, to excite it by an external DC supply two slip rings are enough. One each for positive and negative terminals.

- With the very basic principle of an Alternator, let us now have an insight into its Basic Operational Principle of a **Practical Alternator**.
- During the discussion of the basic working principle of an Alternator, we have considered that the magnetic field is stationary and conductors (armature) is rotating.
- But generally in **practical construction of Alternator :**

***Armature conductors are stationary and Field magnets rotate between them***.

- The **Rotor** of an ***Alternator*** **or** a ***Synchronous generator*** is mechanically coupled to the shaft or the turbine blades, which is made to rotate at **Synchronous speed $N_s$** under some mechanical force **results in magnetic flux cutting** of the **stationary armature conductors housed on the stator.**
- ***As a direct consequence of this flux cutting,*** an induced emf and current starts to flow through the armature conductors which first flow in one direction for the first half cycle and then in the other direction for the second half cycle for each winding with a definite time lag of $120^o$ due to the space displaced arrangement of $120^o$ between them as shown in the figure below.
- ***This particular phenomenon results in three-phase power flow out of the alternator which is then transmitted to the distribution stations for domestic and industrial uses.***

# Voltage Regulation by Synchronous Impedance Method

Voltage regulation of an alternator is the change in terminal voltage from no-load to full-load at constant speed and excitation, expressed as a percentage of rated voltage.

$$\%\text{Voltage Regulation} = \frac{E_0 - V}{V} \times 100$$

Where:

- $E_0$ = No-load induced EMF
- $V$ = Full-load terminal voltage

## Synchronous Impedance (EMF) Method

### Principle

This method assumes that the **armature reaction** and **leakage reactance** can be represented by a single reactance called **synchronous reactance (X_s)**.

$$Z_s = \sqrt{R_a^2 + X_s^2}$$

## Experimental Tests Required

1.Open Circuit Test (OCC) and  Short Circuit Test (SCC)

## Procedure

1. From OCC, find **no-load EMF ($E_0$)** at rated field current.
2. From SCC, find **short-circuit current (I_sc)** at same field current.
3. Calculate synchronous impedance:

$$Z_s = \frac{E_0}{I_{sc}}$$

4. Calculate synchronous reactance:

$$X_s = \sqrt{Z_s^2 - R_a^2}$$

5. Using phasor diagram, find generated EMF:

$$E = \sqrt{(V\cos\phi + IR_a)^2 + (V\sin\phi + IX_s)^2}$$

# Principle of Operation of Synchronous Motor

**Definition**

A synchronous motor is an AC motor that runs at constant synchronous speed irrespective of load.

$N_s = 120f/P$ Where: f = supply frequency P = number of poles

**Principle of Operation**

- When a 3-phase AC supply is given to the stator, it produces a rotating magnetic field.
- DC excitation is applied to the rotor, producing a stationary magnetic field.
- Rotor field locks with stator rotating field due to magnetic attraction.
- Rotor rotates at synchronous speed.

# Stepper Motor – Principle and Applications

**Definition**

**A stepper motor is a special type of motor that converts electrical pulses into precise mechanical movement.**

## Principle of Operation

- Rotor moves in **steps** rather than continuous rotation.
- Each pulse energizes stator windings in sequence.
- Rotor aligns with the energized stator pole.
- Each pulse results in a **fixed angular displacement**.

$$\text{Step Angle} = \frac{360^\circ}{\text{Number of steps per revolution}}$$

# Applications

- **CNC machines**
- **Robotics**
- **Printers**
- **Disk drives**
- **Medical instruments**

# Unit-5

## Electrical Installations and Batteries:

- Basic layout of domestic electrical installation, Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wiring, Earthing: Need of earthing, Types (Pipe and Plate earthing), classification of batteries, working and electrical characteristics of Lead Acid battery, Block diagram of UPS and its operation.

## Low Voltage Switchgear or LV Switchgear

- The apparatus used for switching, controlling and protecting the electrical circuits and equipments is known as SWITCHGEAR.
- Generally electrical switchgear rated upto 1kV is termed as low voltage switchgear.
- LV Switchgear includes low voltage Circuit Breakers (CB), HRC fuses, earth leakage circuit breaker, miniature circuit breakers (MCB) and molded case circuit breakers (MCCB) etc i.e. all the accessories required to protect the LV system.
- The most common use of LV switchgear is in LV distribution board. This system has the following parts



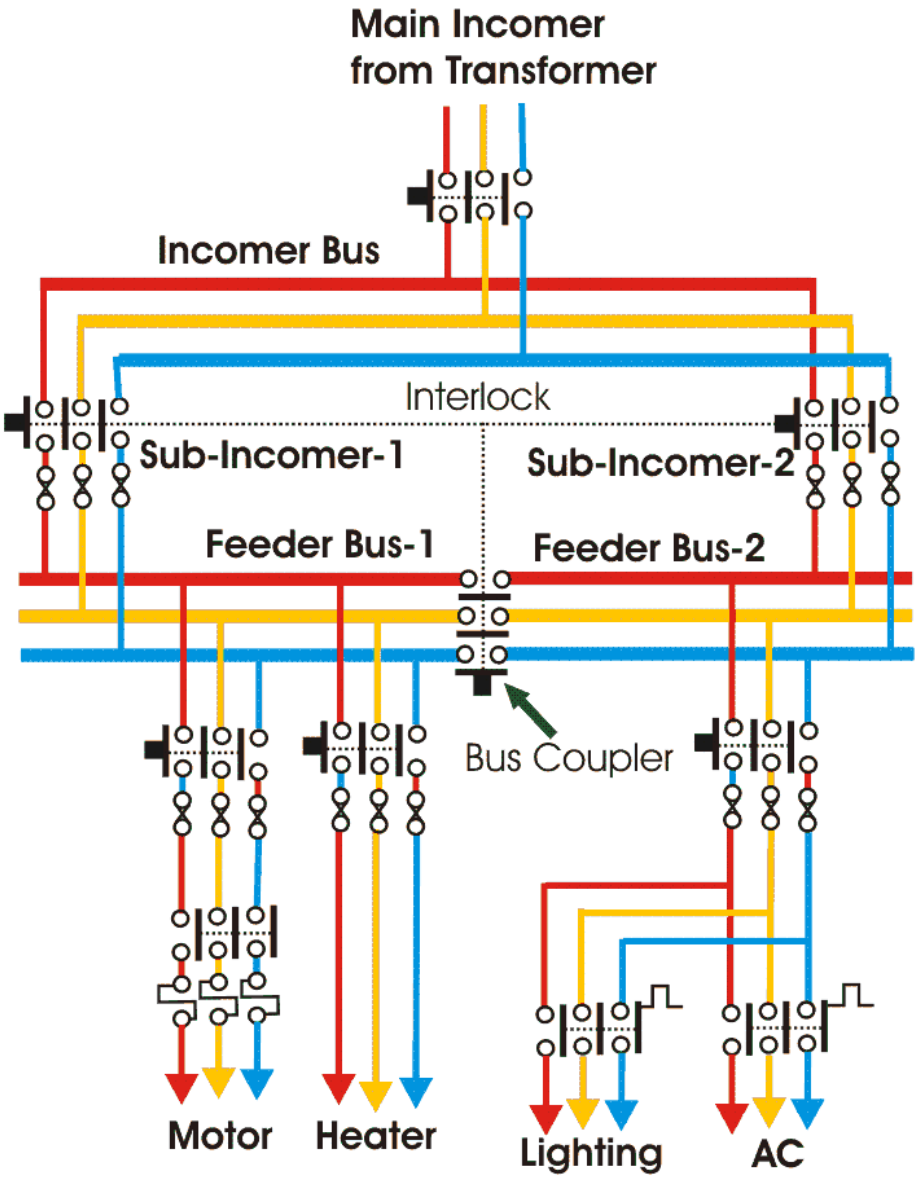
- In **LV switchgear system**, electrical appliances are protected against short circuit and over load conditions by **Electrical Fuses** or **Electrical Circuit Breaker.**

- However, the human operator is not adequately protected against the faults occurs inside the appliances.

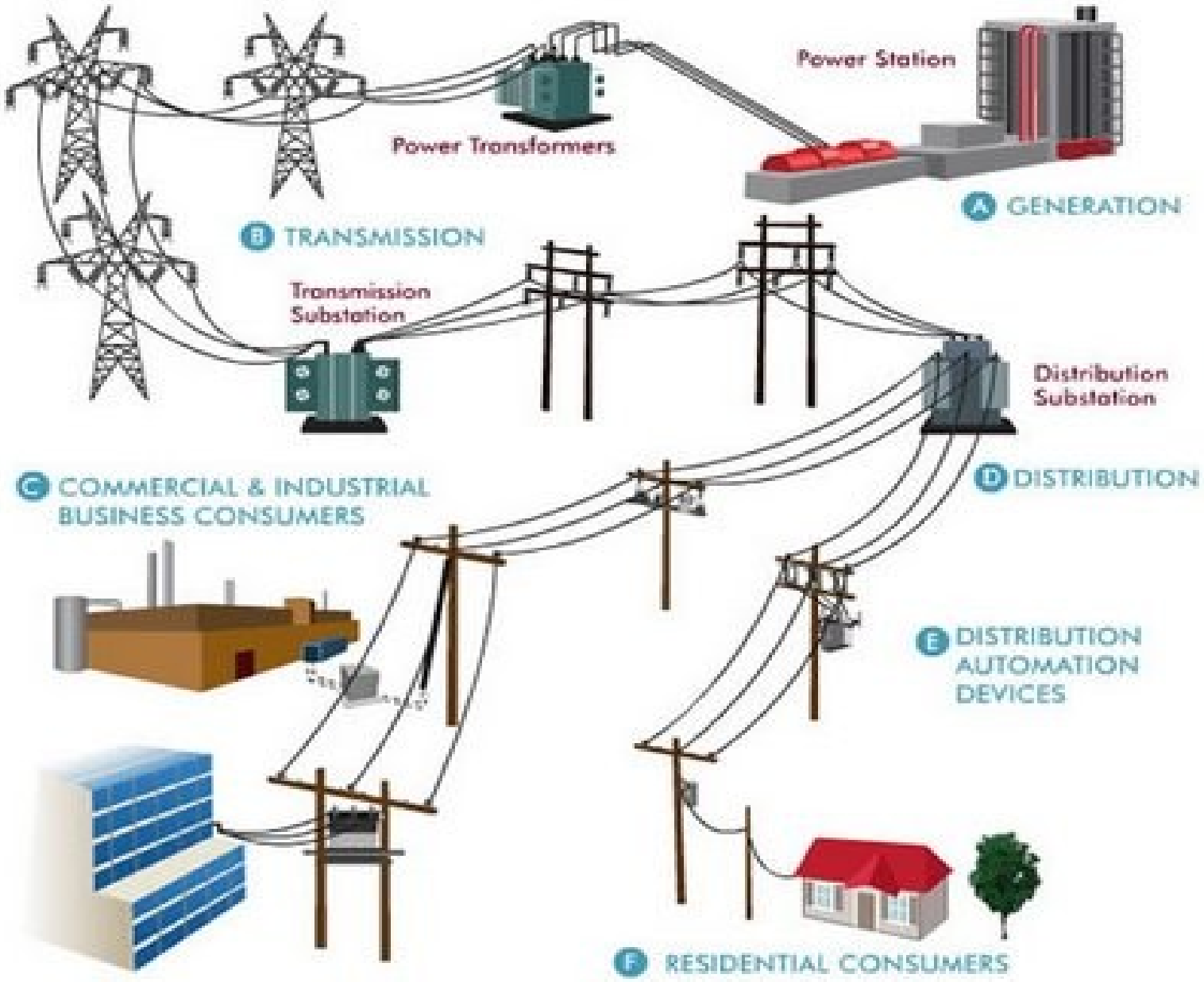
- The problem can be overcome by using earth leakage circuit breaker.

- This ELCB (earth leakage circuit breakers) operates on low leakage current. The earth leakage circuit breaker can detect leakage current as low as 100 mA and is capable of disconnecting the appliance in less than 100 msec.

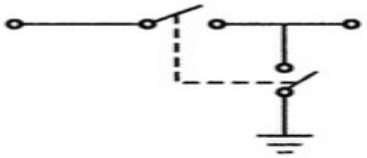
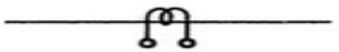
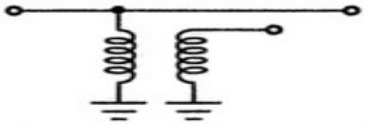
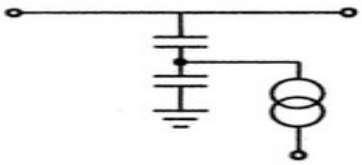
# LV Switchgear



# GENERAL LAYOUT OF POWER SYSTEM:



# Components of LT Switchgear

| Sr. No. | Name of the element | Symbols |
|---|---|---|
| 1. | Bus bar |  |
| 2. | Single break isolating switch |  |
| 3. | On load isolating switch |  |
| 4. | Double break isolating switch |  |
| 5. | Isolating switch with earth blade |  |
| 6. | Current transformer |  |
| 7. | Potential transformer |  |
| 8. | Oil circuit breaker |  |
| 9. | Air blast circuit breaker |  |
| 10. | Air circuit breaker with over current tripping device |  |
| 11. | Capacitive voltage transformer |  |

# Fuse

- **Fuse** is a short piece of metal, inserted in the circuit with low melting point. When high current flows through the circuit due to **overloading** or **short circuit**, the wires gets heated and melts, thus the circuit breaks and current stops flowing.

**Types of Fuses:**

- Rewirable fuse
- HRC (High Rupturing Capacity)

**Rewirable fuse**

- Rewirable fuses are used for low value current i.e upto 200A, it requires **considerable time to replace**

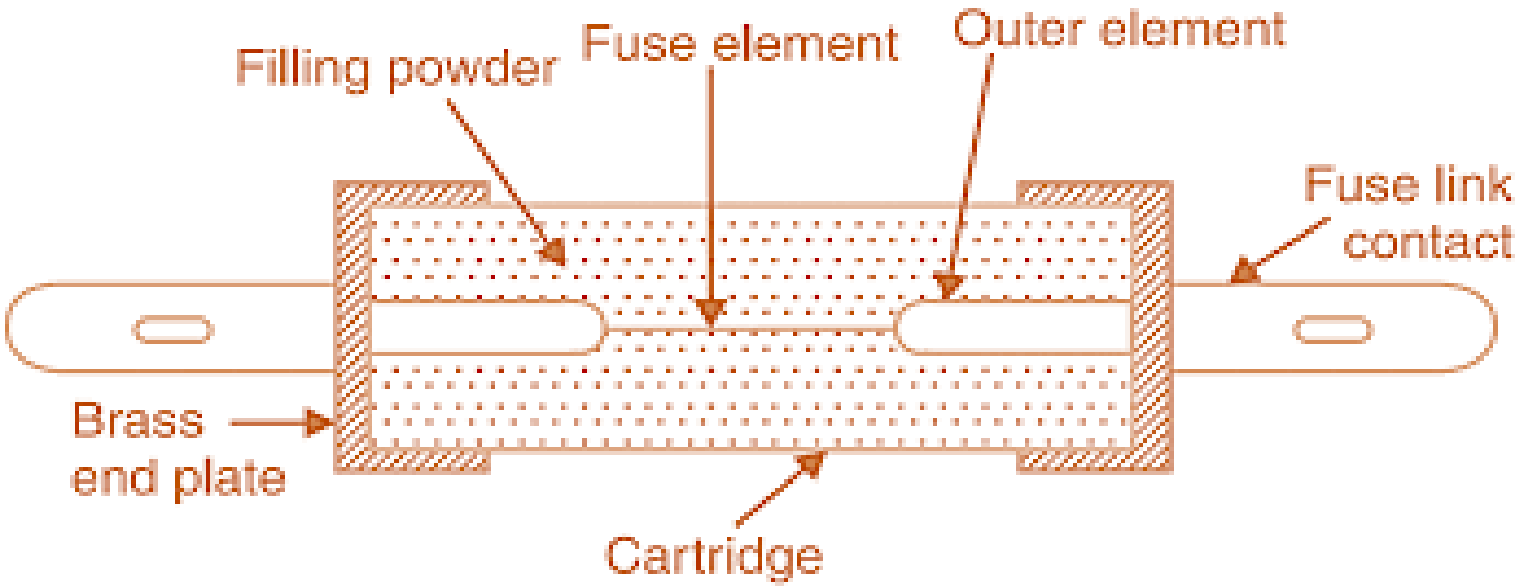


# HRC Fuse

- It consists of certain fuse resistance ceramic body with metal end caps, which are welded to a silver element. The internal space of the body completely filled powder.





# Desirable Characteristics of Fuse element

The material used for fuse wire must have the following characteristics

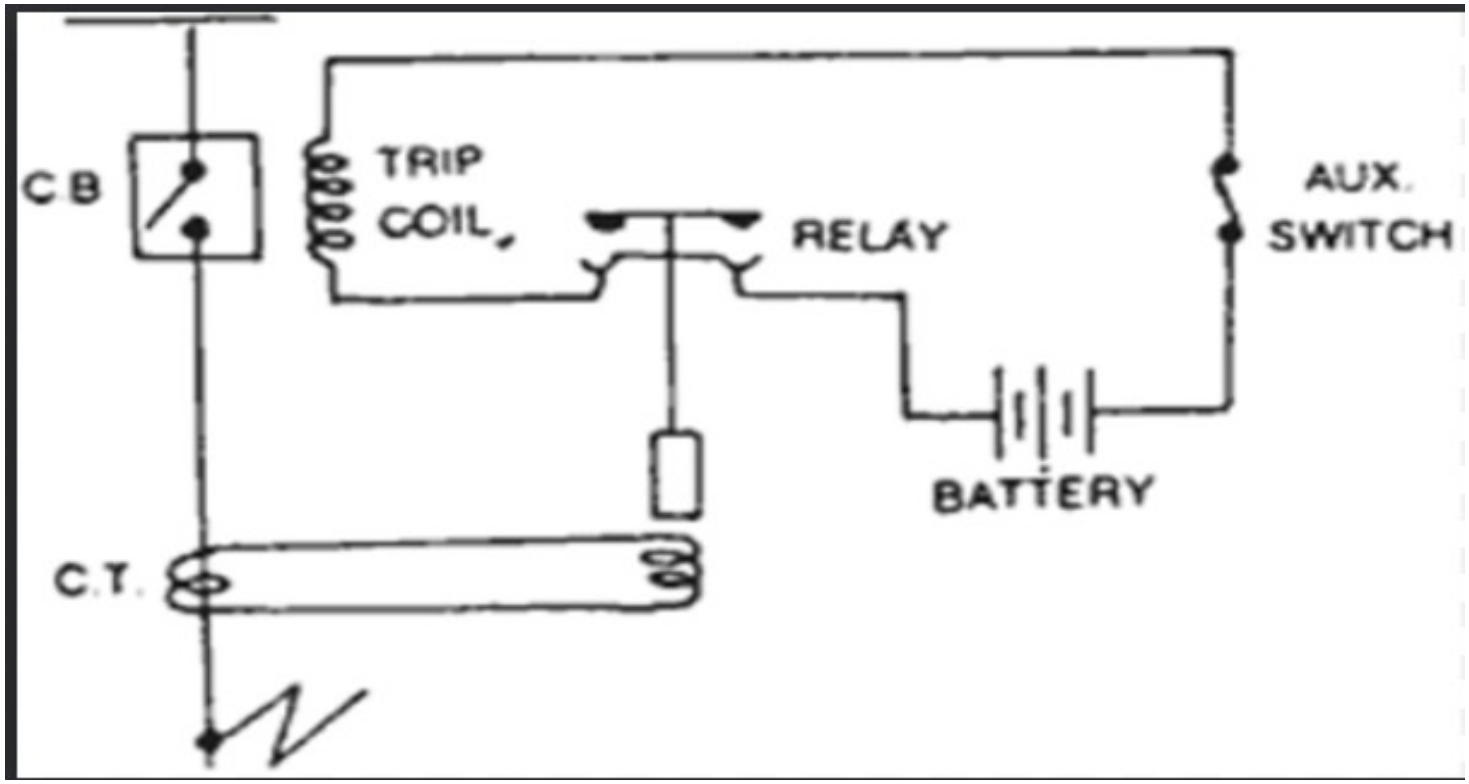
- Low melting point, ex: lead, tin
- High conductivity , ex: Silver, copper
- Free from deterioration due to oxidation e.g. silver.
- Low cost, ex: Lead, tin, copper

# Circuit Breaker (CB)

- **CB** is a switching device which can be operated **manually** and **automatically** for the controlling and protection of electrical power systems respectively.
- During short circuit fault or any other type of electrical fault, these equipments as well as the power network suffers from high stress of fault current, which in turn damages the equipment and networks permanently.
- For saving these equipment and the power networks, the fault current should be cleared form the system as quickly as possible.
- Again after the fault is cleared, the system must come to its normal working condition as soon as possible for supplying reliable quality power to the receiving ends.
- A **CB** essentially consists of **fixed and moving contacts,** called fixed and moving electrodes. Under normal operating conditions, these contacts remains closed, until and unless the system becomes faulty.
- The contacts can be opened manually or by remote control whenever desired. When a fault occurs in any part of the system, the trip coils of the breaker gets energised and the moving contacts are pulled apart by some mechanism, thus opening the circuit.

# Types of CB's

- Miniature circuit breaker (MCB)
- Earth leakage CB (ELCB) or Residual current CB (RCCB)
- Molded Case Circuit Breaker (MCCB)



# Miniature Circuit Breaker (MCB)

- **MCB's** are **electromechanical devices** which protect an electrical circuit from **over currents.**
- Over currents in an electrical circuit may results from short circuit, overload or faulty design.
- MCB is a better alternative than fuse, since it does not require replacement once an overload is detected.
- MCB functions by interrupting the continuity of electrical flow through the circuit once a fault is detected.
- Generally **MCB's** are designed to protect against **over current** and **over temperature faults** (over heating)

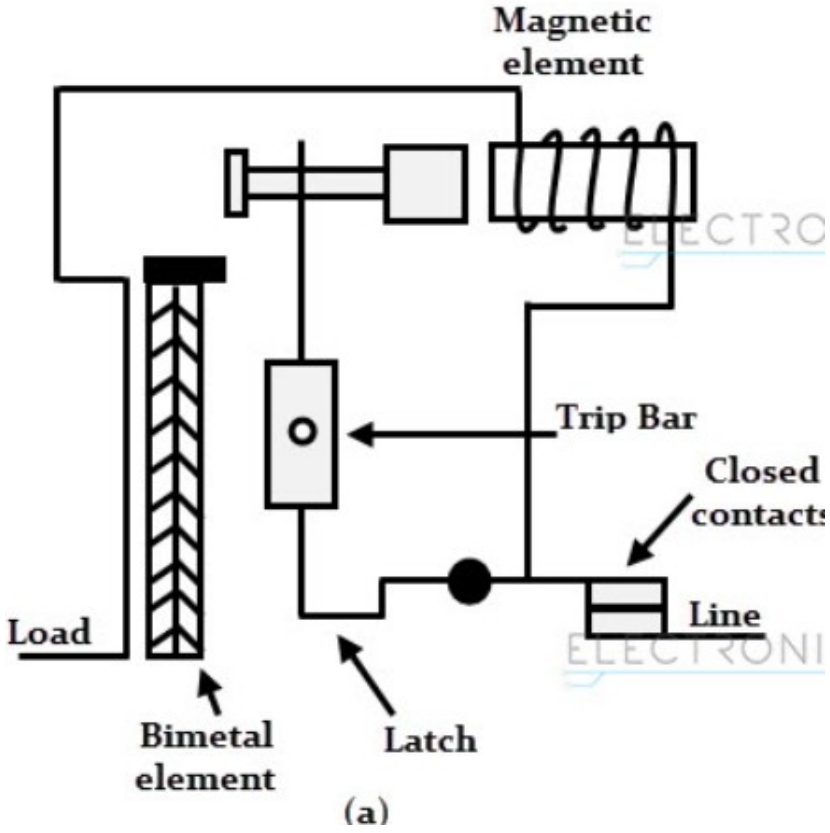
# Working principle

- There are two contacts- one is fixed and the other  is moveable.
- When current exceeds the predefined limit, a solenoid forces the moveable contact to open (i.e to disconnect from the fixed contact) and the MCB turns off, thereby stopping the current from flowing in the circuit.

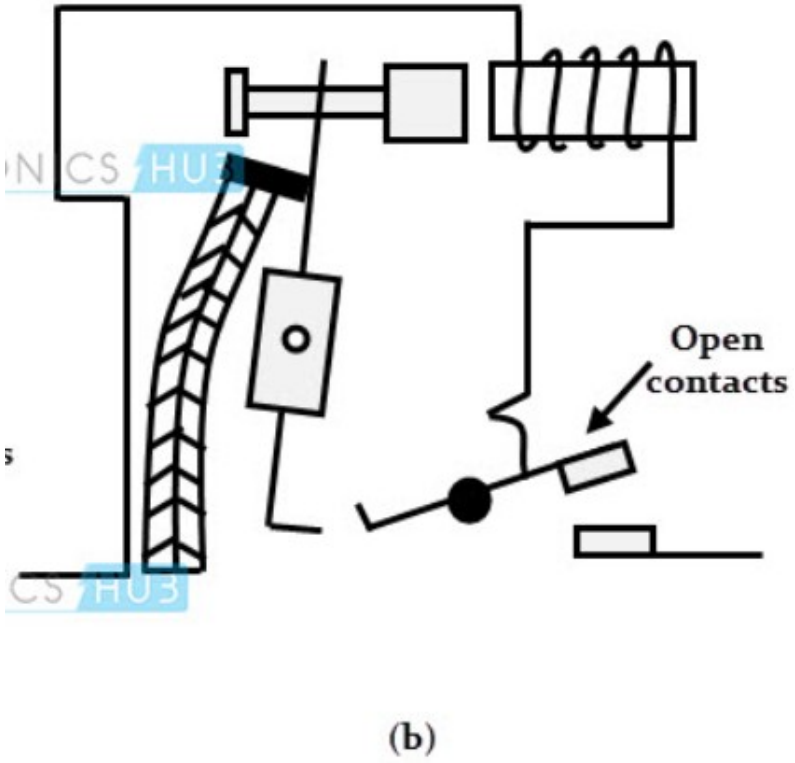
- When **overload** conditions exits, the temperature of bimetallic strip increases as excess current flows through it. Hence due to heat generated in metallic strip the thermal expansion of metal occurs which just creates deflection using deflection of bimetallic strip the latch get removed which makes contacts separate there by protect the circuit.

- Normal operation is shown as follows

## Normal Conditions




## Over Load Condition



# Application of MCB

- Home electrical panels used to prevent from overload or short circuit condition
- Lighting system
- Small scale applications
- Geysers, heaters, AC etc



# Molded Case Circuit Breaker (MCCB)

- **MCCB's** are **electromechnical** devices which protect circuit from **over load, short circuit** conditions. The current ranges from **63A to 200A.**
- These are an alternative to a fuse, since it does not require replacement once an overload is detected. Unlike fuse, an MCCB can be easily reset after a fault and offers improved operational safety and convenience without incurring operating cost.
- MCCB are comprised with five major components such as Molded case (frame), operating mechanism, arc extinguishers, contacts and trip components
- The connection of MCCB for particular application based on specification. MCCB are classified bases on operating mechanisms type of release.

**Thermal magnetic release**

**Electronic release**

**Microprocessor release**

- **Thermal magnetic release:** This type **uses bimetallic strip to protect devices.** These MCCB will have time current characteristics. Tripping characteristics depending upon strip amp temperature. It is not frequently used in electrical installations.

- **Electronic release**: In this case a circuit is placed on power device and **used to protect devices from over current situation**. Since operating characteristics does not depends upon temperature.

- **Microprocessor release:** In this case, microprocessor is **used to monitor over current** in circuit if current exceeds it determining value, The circuit breaker isolates circuit from faulty conditions. This type is used and has several advantages like accuracy reliability when compared with other MCCB.

## Applications:

- MCCB’s can have very high current ratings, which allows them to be used in heavy duty applications
- such as main
- Electric Feeder Protection, Capacitor Bank protection, Generator Protection,
- Welding applications,
- low current applications that requires adjustable trip settings and motor protections.



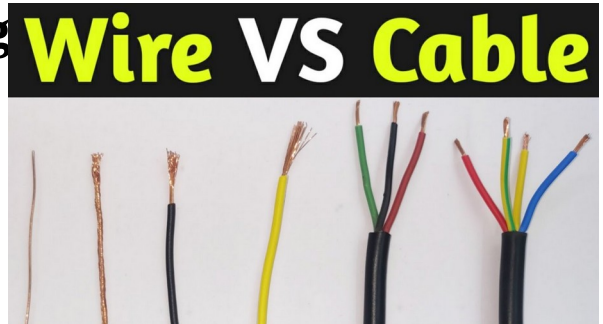
| MCB | MCCB |
| --- | --- |
| Helps in protecting the circuit from overload conditions | Helps in protecting the equipment from over current and short circuit conditions |
| Possess a fixed tripping circuit | Possess a moveable tripping circuit |
| One of three pole configuration is possible | One of four pole configuration is possible |
| The current which can be interrupted is 1800A | The current which can be interrupted is 10 KA – 200 KA |
| Remote operation is not possible | Remote operation is possible |
| The rated current of the device is 100A | The rated current is 10A to 200A |
| Can be installed for light loads | Can be installed for heavy loads |
| | |
| | |

# Difference between Wire and Cable

- **According to Bureau of Indian Standards (*BIS*) , wire and cable can be defined as follows:**
- **Wire**: If a bare conductor is provided with insulation, then it is known as a wire. The insulation separates the conductor electrically from other conductor.
- **Cable**: It consists of two or more conductors covered with suitable insulation and surrounded by a protecting cover.

**The cable essentially consists of three parts:**

- **Conductor or core** - the metal wire, or strand of wires, carrying the current
- **Insulation or dielectric** - a covering of insulating material to avoid leakage of currents from the conductor and
- **Protective covering** from mechanical damage



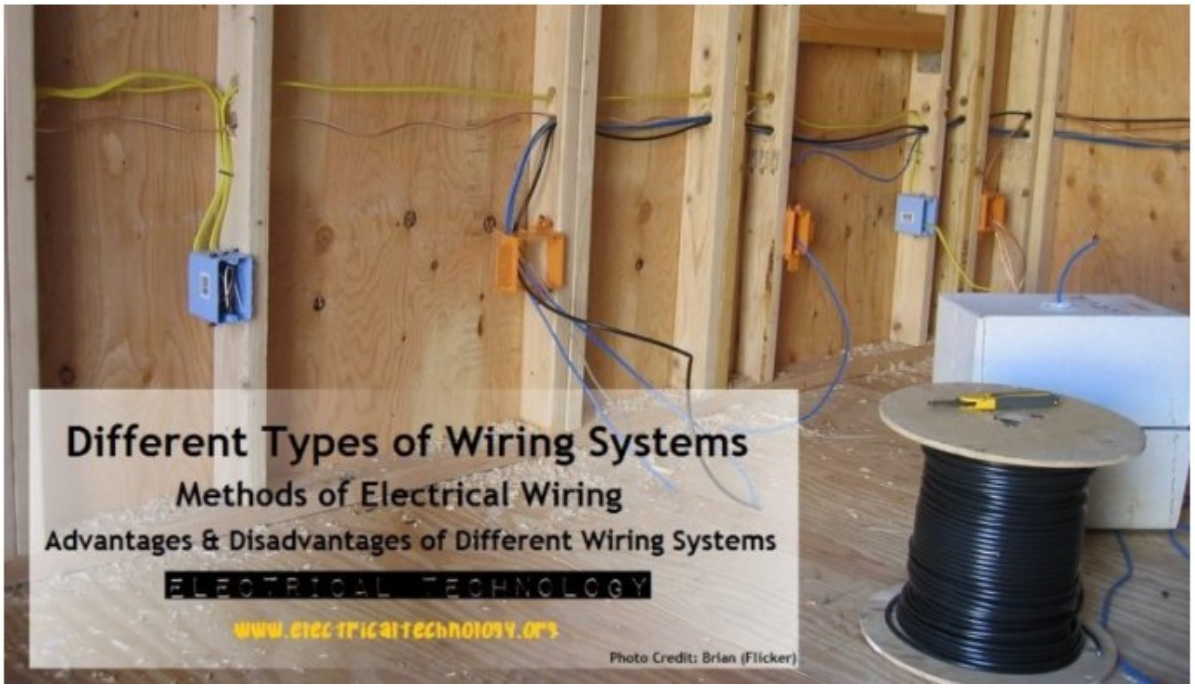
# Types of Electrical Wiring

Electrical wiring system can be classified into the following five categories:

1. **Cleat wiring**
2. **Casing and Capping wiring**
3. **Batten wiring (CTS or TRS)**
4. **Conduit wiring**

## Cleat Wiring

- This system of wiring comprise of ordinary PVC insulated wires (occasionally, sheathed and weather proof cable) braided and compounded held on **walls or ceilings** by means of porcelain cleats, Plastic or wood.
- Cleat wiring system is a **temporary wiring system** therefore it is not suitable for domestic premises. The use of cleat wiring system is over nowadays.



# Advantages of Cleat Wiring:

- It is simple and cheap wiring system
- Most suitable for temporary use i.e. under construction building or army camping
- As the cables and wires of cleat wiring system is in open air, Therefore faul t in cables can be seen and repair easily.
- Cleat wiring system installation is easy and simple.
- Customization can be easily done in this wiring system e.g. alteration and addition.
- Inspection is easy and simple.

# Disadvantages of Cleat Wiring:

- Appearance is not so good.
- Cleat wiring can’t be use for permanent use because, Sag may be occur after sometime of the usage.
- In this wiring system, the cables and wiring is in open air, therefore, oil, Steam, humidity, smoke, rain, chemical and acidic effect may damage the cables and wires.

- it is not lasting wire system because of the weather effect , risk of fire and wear & tear.
- it can be only used on 250/440 Volts on low temperature.
- There is always a risk of fire and electric shock.
- it can’t be used in important and sensitive location and places.
- It is not lasting, reliable and sustainable wiring system.

# Casing and Capping wiring

- Casing and Capping wiring system was famous wiring system in the past but, it is considered obsolete this days because of Conduit and sheathed wiring system. The cables used in this kind of wiring were either VIR or PVC or any other approved insulated cables.

- The cables were carried through the wooden casing enclosures. The casing is made up of a strip of wood with parallel grooves cut length wise so as to accommodate VIR cables. The grooves were made to separate opposite polarity. the capping (also made of wood) used to cover the wires and cables installed and fitted in the casing.


www.electricaltechnology.org


Casing Capping Joints


Casing & Capping Wiring

# Advantages of Casing Capping Wiring:

- It is **cheap wiring system** as compared to sheathed and conduit wiring systems.
- It is **strong** and **long-lasting** wiring system.
- Customization can be easily done in this wiring system.
- If Phase and Neutral wire is installed in separate slots, then repairing is easy.
- Stay for long time in the field due to strong insulation of capping and casing..
- It stays safe from oil, Steam, smoke and rain.
- No risk of electric shock due to covered wires and cables in casing & capping.

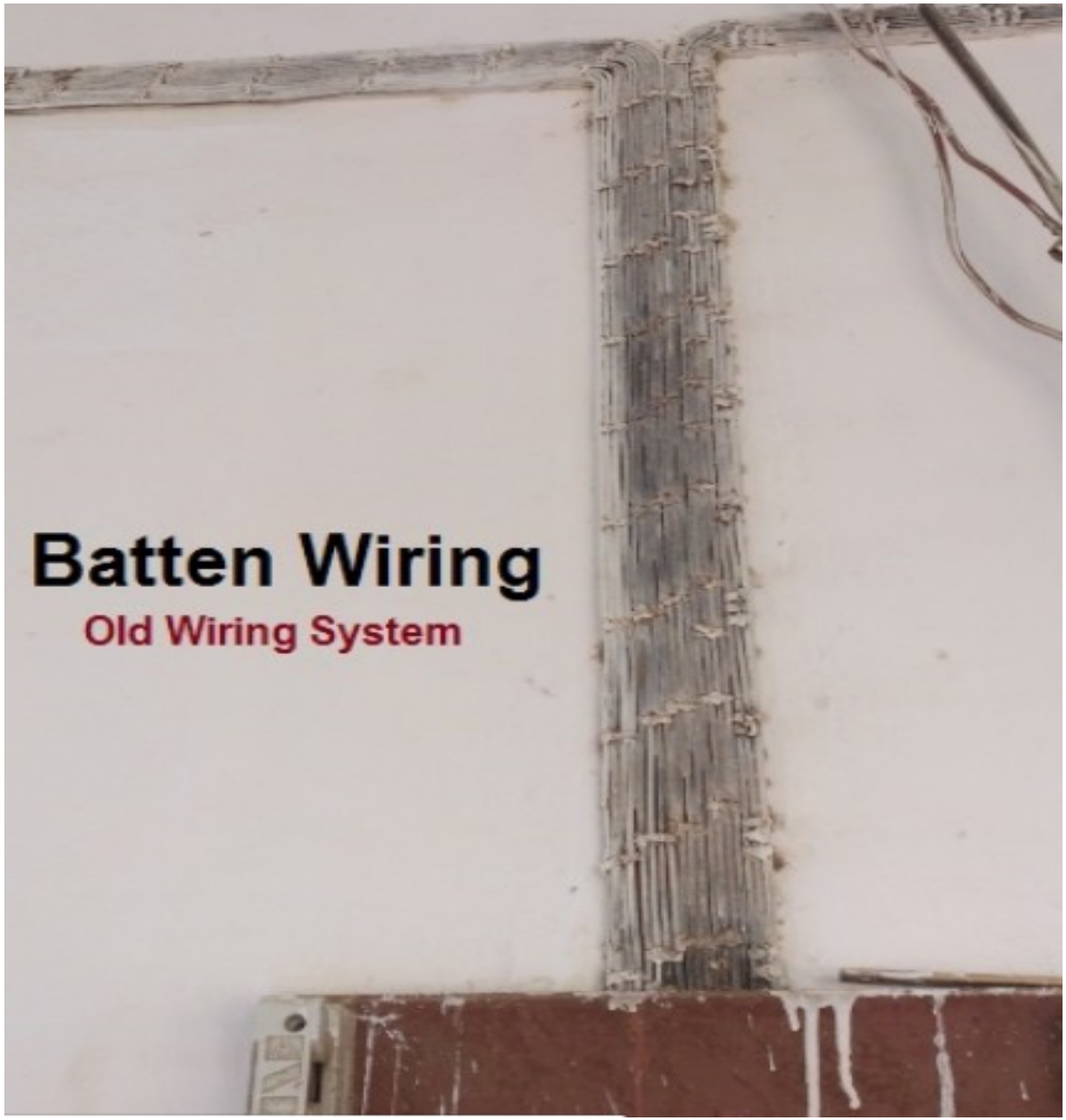
# Disadvantages Casing Capping Wiring:

- There is a high risk of fire in casing & capping wiring system.
- Not suitable in the acidic, alkalies and humidity conditions
- Costly repairing and need more material.
- Material can’t be found easily in the contemporary
- White ants may damage the casing & capping of wood.

# Batten Wiring (CTS or TRS)

- **Single core or double core or three core TRS cables with a circular oval shape cables are used in this kind of wiring**. Mostly, single core cables are preferred. TRS cables are chemical proof, water proof, steam proof, but are slightly affected by lubricating oil. The TRS cables are run on well seasoned and straight teak wood batten with at least a thickness of 10mm.

- The cables are held on the wooden batten by means of tinned brass link clips (buckle clip) already fixed on the batten with brass pins and spaced at an interval of 10cm for horizontal runs and 15cm for vertical runs.


Batten Wiring
Old Wiring System

# Advantages of Batten Wiring

- Wiring installation is simple and easy
- Cheap as compared to other electrical wiring systems
- Paraphrase is good and beautiful
- Repairing is easy
- Strong and long-lasting
- Customization can be easily done in this wiring system.
- Less chance of leakage current in batten wiring system

# Disadvantages of Batten Wiring

- Can’t be install in the humidity, Chemical effects, open and outdoor areas.
- High risk of fires
- Not safe from external wear & tear and weather effects (because, the wires are openly visible to heat, dust, steam and smoke.
- Heavy wires can’t be used in batten wiring system.
- Only suitable below thasn 250V.
- Need more cables and wires.

# Conduit Wiring







Surface Conduit Wiring






Concealed Conduit wiring

## Advantage of Conduit Wiring Systems

- It is the safest wiring system (Concealed conduit wring)
- Appearance is very beautiful (in case of concealed conduit wiring)
- No risk of mechanical wear & tear and fire in case of metallic pipes.
- Customization can be easily done according to the future needs.
- Repairing and maintenance is easy.
- There is no risk of damage the cables insulation.
- It is safe from corrosion (in case of PVC conduit) and risk of fire.
- It can be used even in humidity , chemical effect and smoky areas.
- No risk of electric shock (In case of proper earthing and grounding of metallic pipes).
- It is reliable and popular wiring system.
- Sustainable and long-lasting wiring system.

# Disadvantages of Conduit Wiring Systems

- It is expensive wiring system (Due to PVC and Metallic pipes, Additional earthing for metallic pipes Tee(s) and elbows etc.
- Very hard to find the defects in the wiring.
- Installation is not easy and simple.
- Risk of Electric shock (In case of metallic pipes without proper earthing system)
- Very complicated to manage additional connection in the future.

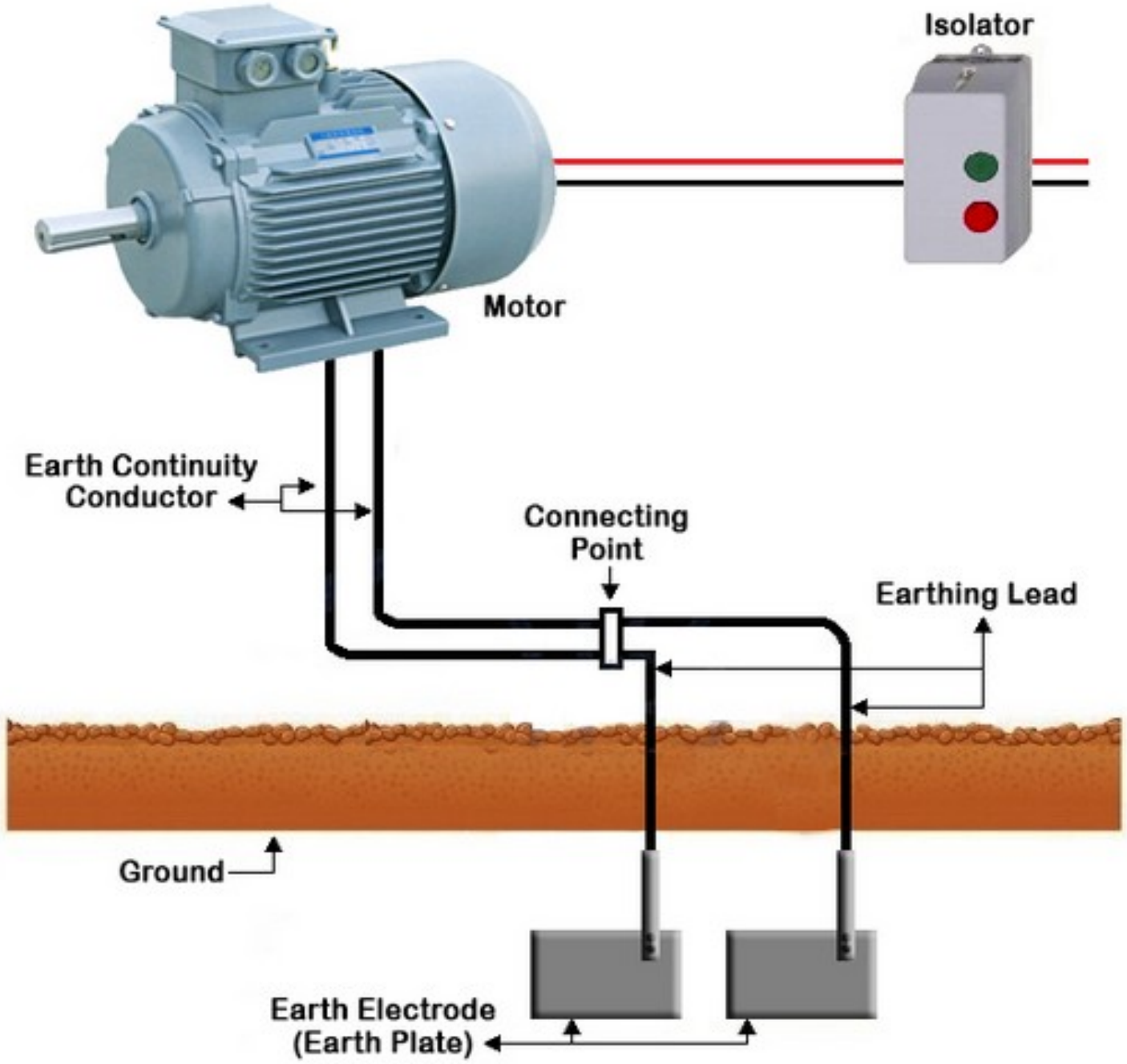
# Earthing

- The process of connecting the metallic frame (i.e non-current carrying part) of electrical equipment or some electrical part of the system (e.g neutral point in a star connected system or one conductor of the secondary of a transformer etc) to the earth (i.e soil) is called **Earthing or Grounding.**

- **Earthing** and **Grounding** is the same terms used for earthing. **Grounding is the commonly word** used for earthing in the **North American** standards like **IEEE, NEC, ANSI** and **UL** etc while, **Earthing is used in European**, Common wealth countries and ***Britain standards like IS and IEC*** etc.

## Necessity of Earthing:

- To protect the operating personnel from the danger of shock
- To maintain the line voltage constant, under unbalanced load conditions
- To avoid risk of fire due to earth leakage current through unwanted path
- For protection of equipment
- For protection of large building and all machines fed from overhead lines against


Isolator
Motor
Earth Continuity
Conductor
Connecting
Point
Earthing Lead
Ground
Earth Electrode
(Earth Plate)

# Earthing serves several important purposes in electrical systems:

- Safety: Earthing minimizes the risk of electric shocks by diverting fault currents away from people and objects.
- Equipment Protection: It helps prevent damage to electrical equipment and appliances by providing a low-resistance path for fault currents.
- Lightning Protection: Proper earthing reduces the risk of lightning strikes by providing a path for lightning currents to safely dissipate into the ground.
- Electromagnetic Interference (EMI) Reduction: Earthing helps mitigate electromagnetic interference, ensuring the proper functioning of sensitive electronic devices.

# Methods of Earthing

- **The various methods of earthing in common use are:**

**1.) Pipe Earthing**

**2.) Plate Earthing**

1.Pipe earthing is the best and most efficient way of earthing and is also easily affordable. Pipe earthing uses 38mm diametre and 2 metres length pipe vertically embedded in the ground to work as earth electrodes.

2.In plate earthing, an earthing plate made of copper or G.I. is buried into the ground at a depth more than 3 metres from the ground level. This earthing plate is embedded in an alternative layer of coke and salts.

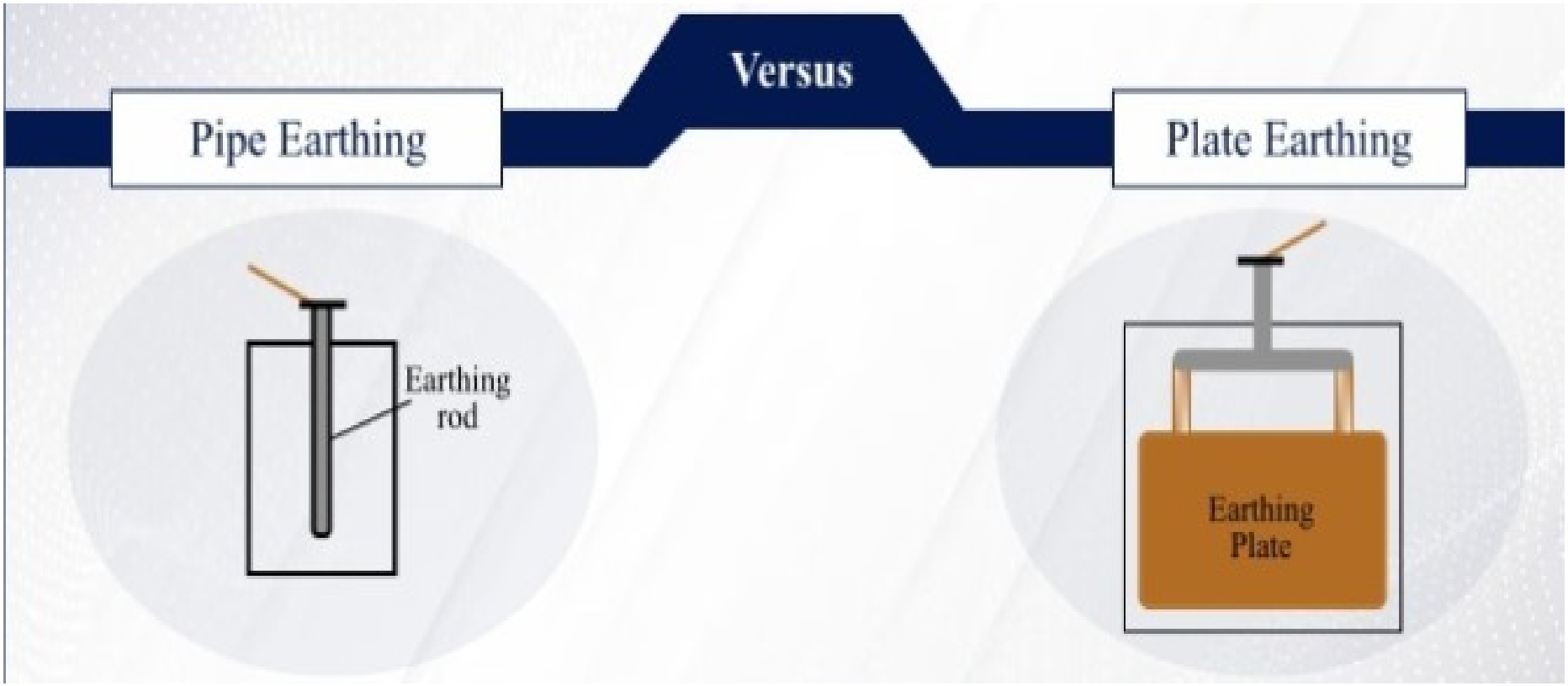


*Copper strip used as Earthign Lead*

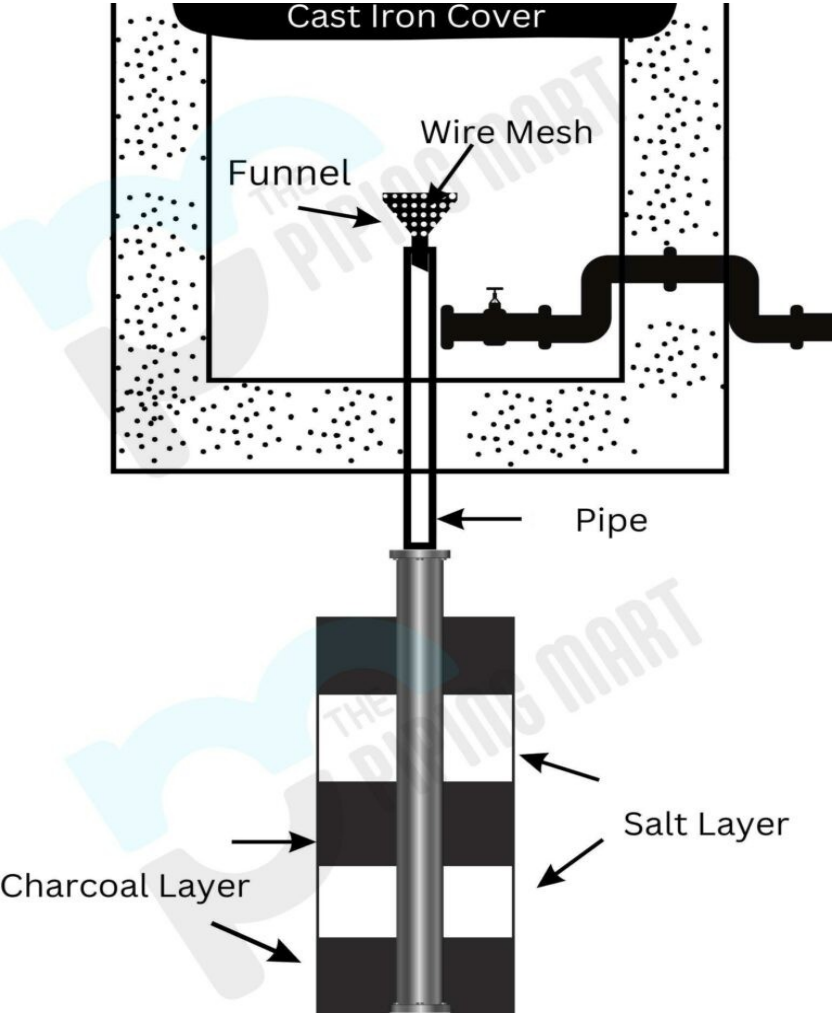


*Galvanised iron strip used as Earthing Lead*

The selection of earthing between pipe earthing and plate earthing is made based on soil condition, current carrying capacity, installation space, cost, etc.
The most significant difference between plate earthing and pipe earthing is in their type of earthing electrode used, i.e. pipe earthing uses a metal rod or pipe as an earthing electrode, while plate earthing uses a metal plate as an earthing electrode.



## Pipe Earthing




## Plate Earthing

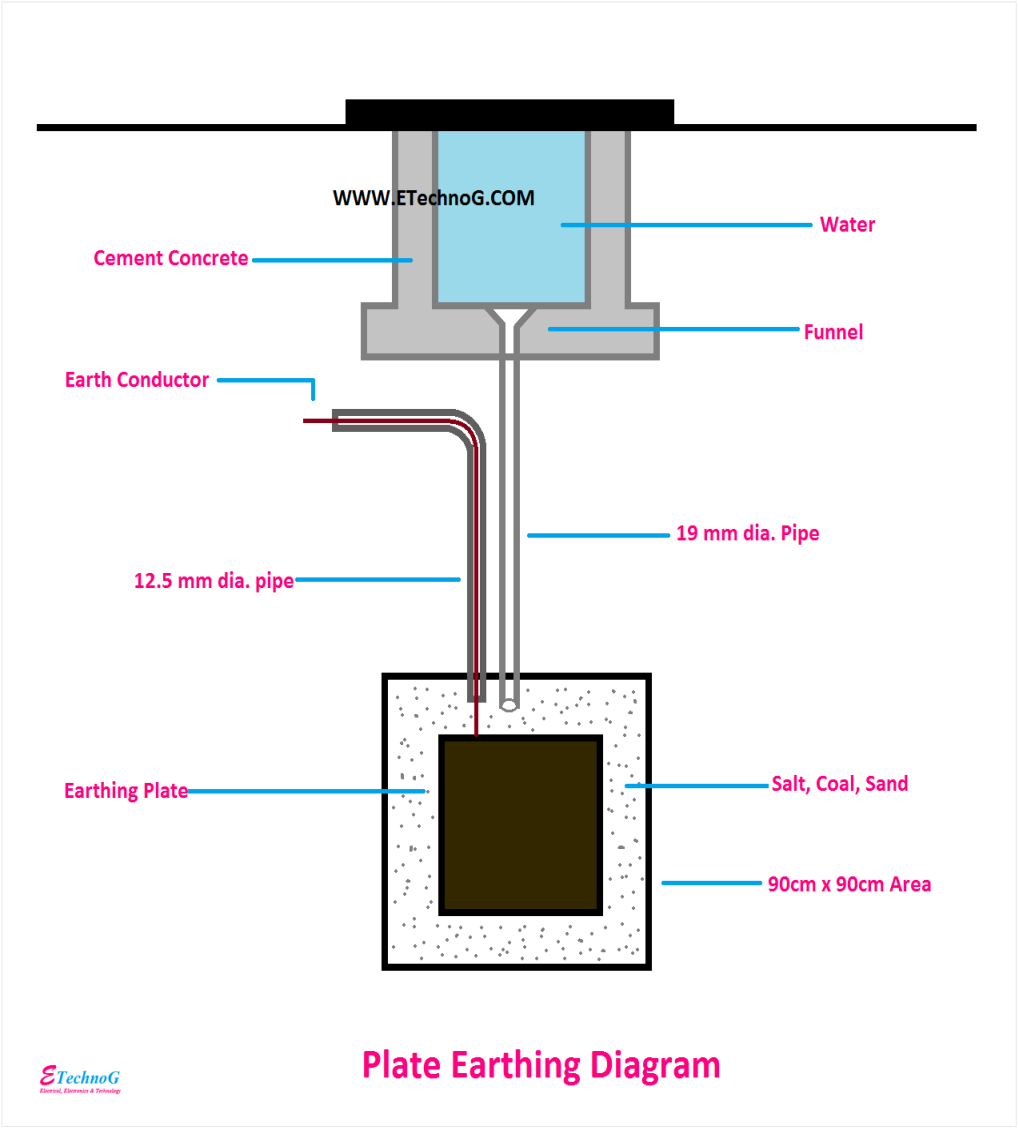



Plate Earthing Diagram

# Pipe Earthing

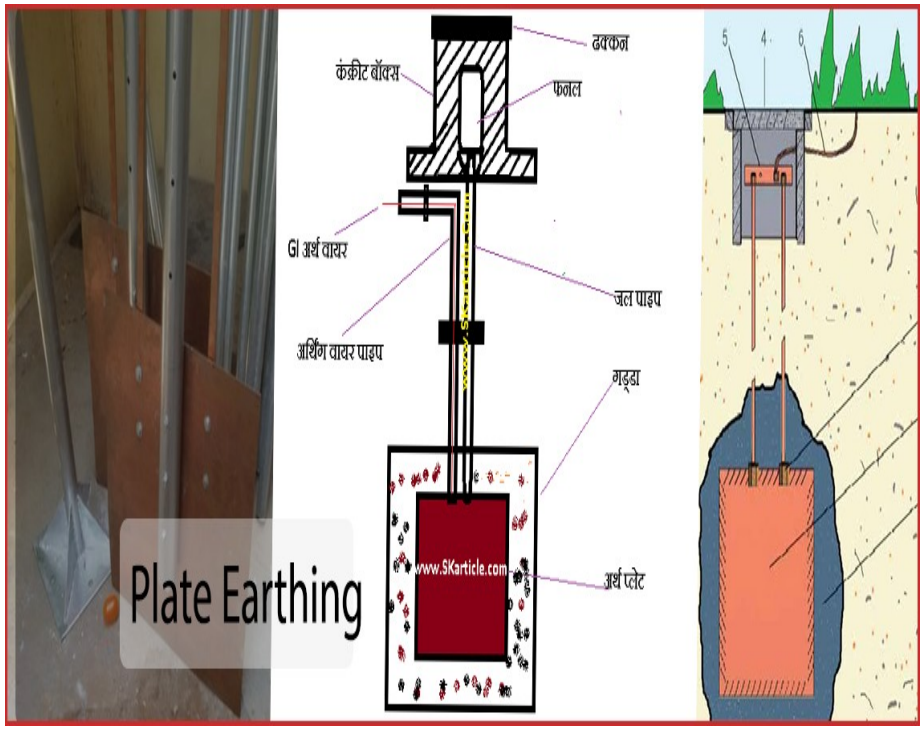


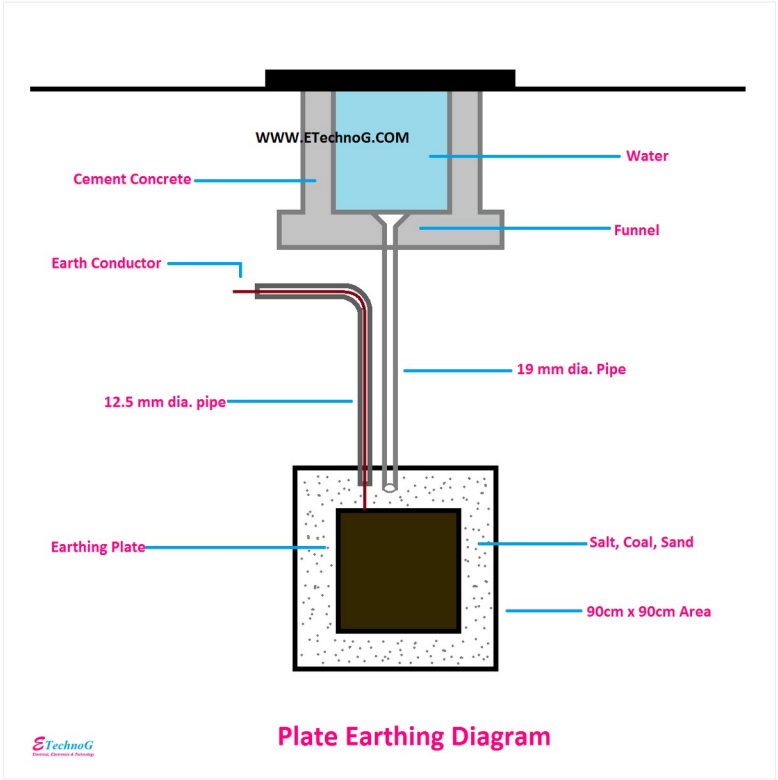



- Pipe earthing is a type of electrical earthing in which an earthing rod made up of copper-coated steel is buried into the ground.
- The other end of the rod is connected to the electrical equipment or system through a conductor wire.
- The rod is typically buried into the ground at least 8 feet deep to ensure proper earthing with maximum efficiency.

- Pipe earthing is also known as rod earthing, as it use a pipe or road as the earth electrode.
- The pipe earthing is known for its lower cost and good corrosion resistance than other types of earthing systems.
- Another major advantages of pipe earthing include quick and easy installation, and less maintenance requirement. Pipe earthing is mainly employed where the soil has low resistivity and high moisture content.

# Plate Earthing



- Plate earthing is another commonly used type of electrical earthing system employed in electrical systems to provide protection against electrical accidents. In plate earthing system, a flat metal plate is buried into the ground which acts as an earth electrode in the earthing system.
- This metal plate is typically made up of galvanized iron (GI) or copper. The metal is usually buried into the ground at least 4 feet deep to ensure maximum earthing efficiency.

- This metal plate is then connected to the electrical system through a highly conductive wire. The size of the metal is determined according to the soil resistance and current-capacity required.

- Plate earthing is mostly used in areas where soil resistance is relatively high and moisture content is low, such as in dry or rocky soil areas.

- After getting a brief overview of pipe earthing and plate earthing, let us now discuss the important differences between them.

**Advantages:**

- Tried and tested method for earthing
- Economic and easily available

**Disadvantages:**

- Salts get washed away during rains or water flow
- Plate gets corroded and thus has a short life span
- Due to high current flow, coal may burn and get reduced into ash, thus creating resistance instead of being conductive
- Regular addition of water is must.
- Plate earthing is now-a-days being replaced by Advance Maintenance Free Earthing (pipe in pipe/ Strip in pipe/ Solid Copper Bonded Earthing Rods)

# Batteries

- A **Battery** is a device which converts **chemical energy** into **electrical energy** and is **made up of number of cells**. Batteries consists of two or more **voltaic cells** that are connected in **series** to **provide** a **steady DC voltage** at the battery's output terminals, the voltage produced by chemical reaction inside the cell. **Electrodes are immersed in an electrolyte**, which forces the electric charge to separate in the form of ions and free electrons.
- A battery's **Voltage Output** and **Current Rating** are determined by the elements used for the electrodes, the size of the electrodes, and the type of electrolyte used.
- Whether battery may be recharged or not, depends on the cells used to make up the battery.
- **Batteries are classified into two types such as**

**Primary Batteries**

**Secondary Batteries**

**Voltage : The theoretical voltage of the battery is determined by:** $\mathbf{E_{Battery} = E_{cathode} - E_{anode}}$

# Primary Batteries

- As the name indicates, these Batteries are meant for single usage.
- Once these batteries are used they cannot be recharged as the devices are not easily reversible and active materiel may not return to their original forms.
- Other name for these batteries is disposable batteries, some of the examples for disposable batteries are the normal AA, AAA batteries which we used in **wall clock, television remotes** etc.











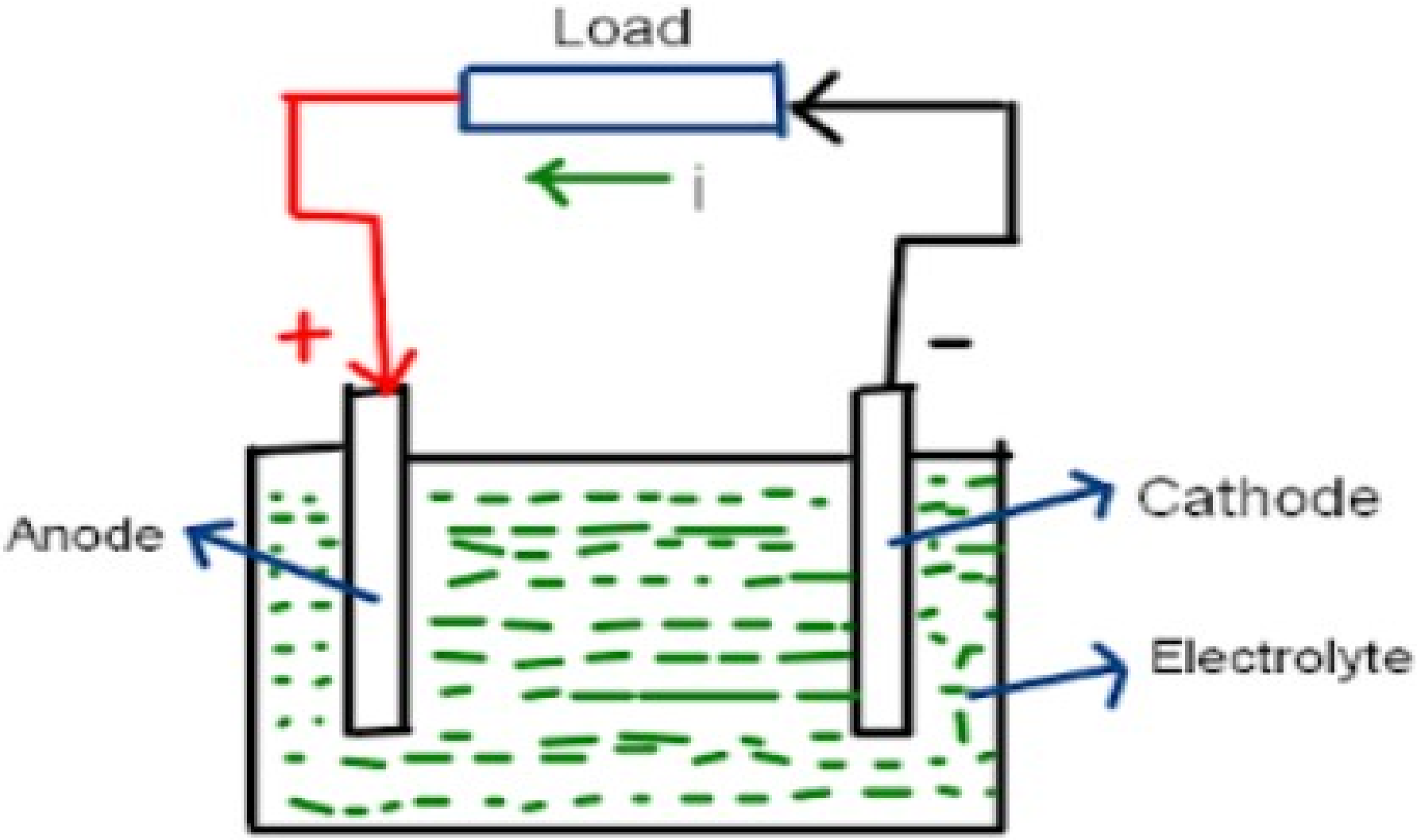

# Secondary Batteries

- Secondary batteries are also known as Rechargeable batteries. These batteries can be used and charged simultaneously.
- A secondary battery or storage battery can be recharged because its chemical reaction is reversible.
- Rechargeable batteries are (re)charged by applying electric current, which reverses the chemical reactions that occur during discharge/use.
- Some of the examples for rechargeable batteries are the batteries used in **mobile phones, MP3 players** etc.
- There are three commonly used secondary batteries in the industry namely,

- ***Lead Acid***
- ***Nickel- iron-alkaline cell***
- ***Nickel-cadmium-alkaline cell***


EXIDE
INVAREO
TUBULAR
EXIDE
EXIDE
EXIDE
EXIDE
EXIDE
EXIDE
Eternity
EXIDE
JAI KISAN
EXIDE

| Primary Cell | Secondary Cell |
| --- | --- |
| If discharged once cannot be charged again | If discharged once can be charged again |
| Light in weight | Heavy in weight |
| Used for intermittent use with low load current rating | Used for continuous rating with high load current rating |
| Short life | Long Life |
| Low cost | High Cost |
| Low efficiency | High efficiency |
| Low power output | High power output |
| Less maintenance | More maintenance |




**Battery**

# Lead Acid Battery Operation

- A lead acid battery consists of a negative electrode made of spongy or porous lead. The lead is porous to facilitate the formation and dissolution of lead. The positive electrode consists of lead oxide. Both electrodes are immersed in a electrolytic solution of sulfuric acid and water. In case the electrodes come into contact with each other through physical movement of the battery or through changes in thickness of the electrodes, an electrically insulating, but chemically permeable membrane separates the two electrodes. This membrane also prevents electrical shorting through the electrolyte. Lead acid batteries store energy by the reversible chemical reaction shown below
- The overall chemical reaction is

$$PbO_2 + Pb + 2H_2SO_4 \underset{charge}{\overset{discharge}{\Longleftrightarrow}} 2PbSO_4 + 2H_2O$$

Positive plate: Lead dioxide ($PbO_2$)

Negative plate: Spongy lead (Pb)

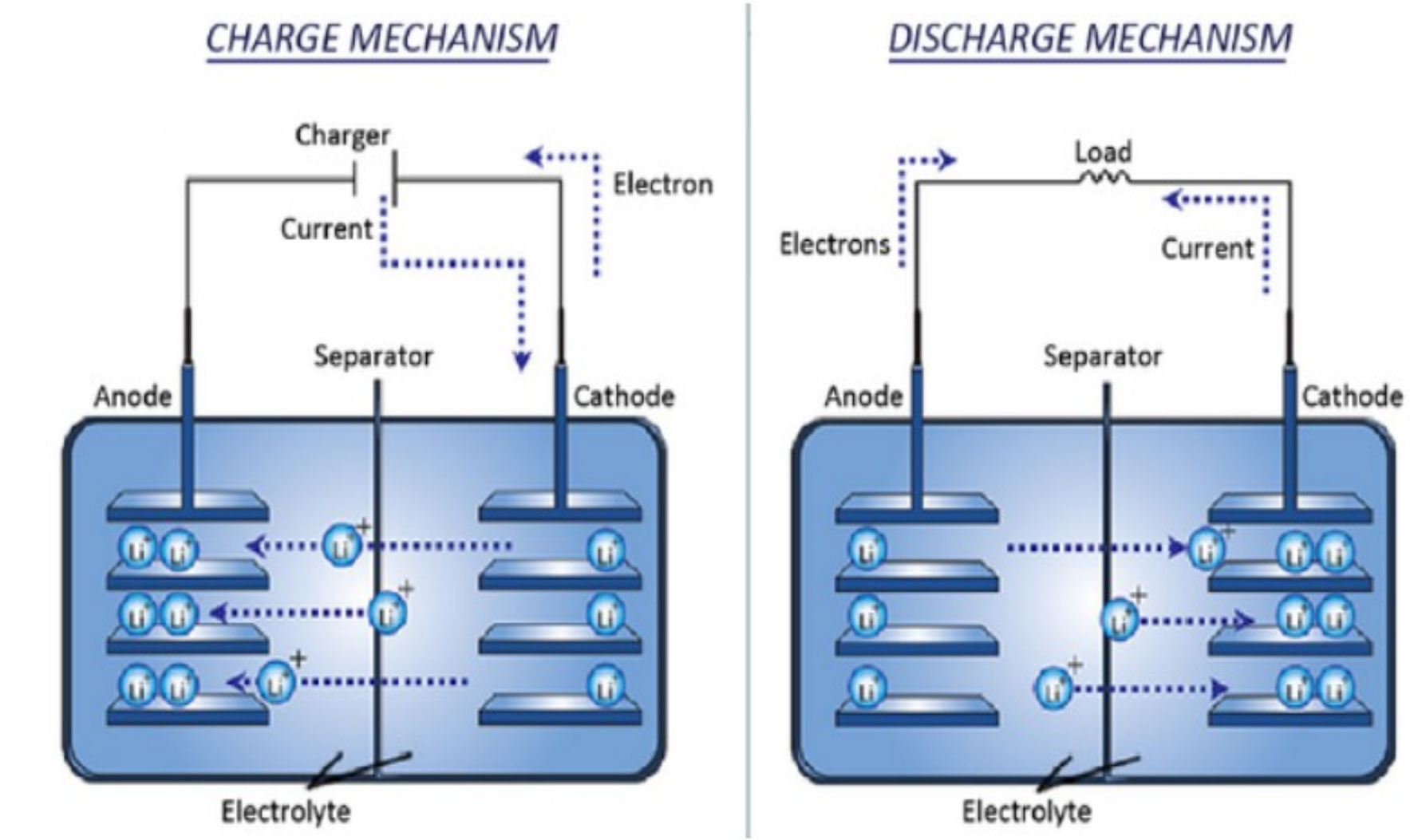
Electrolyte: Dilute sulphuric acid ($H_2SO_4$)

# Important Characteristics of a Lead-Acid Cell

- **Terminal Voltage** - When the battery delivers current, the voltage terminal voltage is less than its EMF due to its internal resistance. Lead acid cell has less lead sulphate that will clogged the pores of the battery once there is continuous flow of current.
- **EMF**- The emf of a fully charged L-A cell is relatively higher, its open circuit voltage will reach as high as 2.2 Volts while other has 1.2 Volts only.
- **Capacity** - The capacity of the cell is defined as the quantity of electricity which it can give out during single discharge until its terminal voltage falls to 1.8 V. Battery capacity is measured by Ampere-hours and the capacity of lead-acid cell is not permitted to discharged beyond 1.8 V, thus it has high capacity.

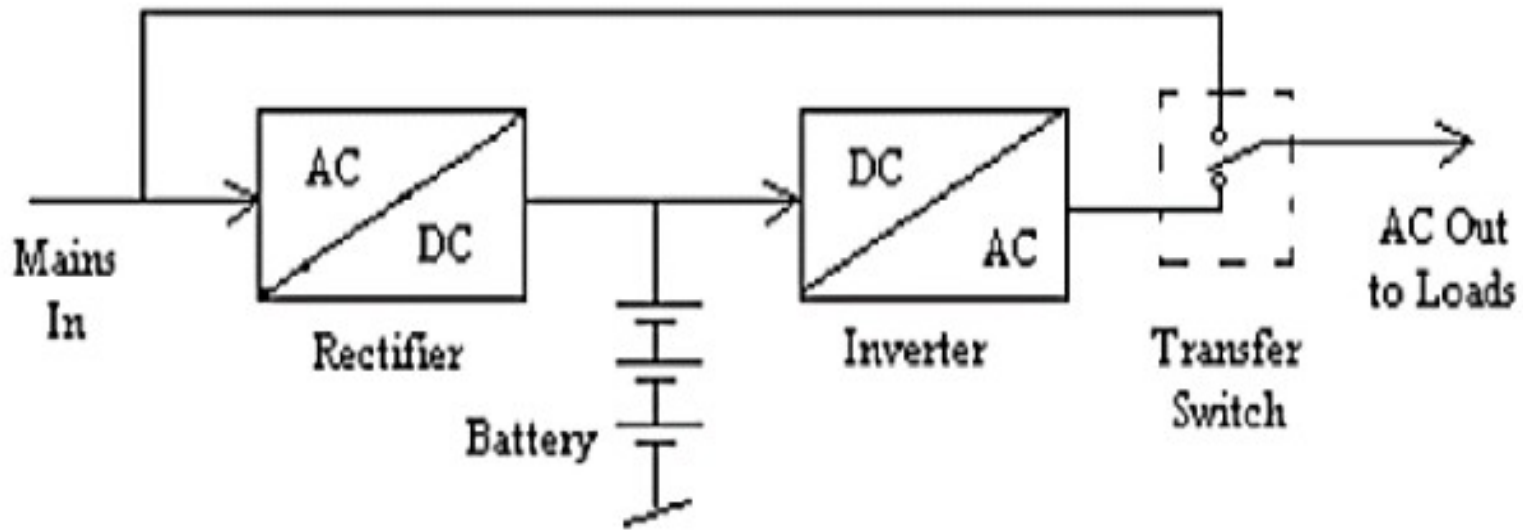
  **Efficiency** - There are two ways which we can measure the efficiency of the lead-acid cell that is ampere-hour efficiency and watt-hour efficiency. The L-A cell has 90% Ampere-hour efficiency and 75% Watt-hour efficiency. These data is relatively higher compare to other secondary cells.

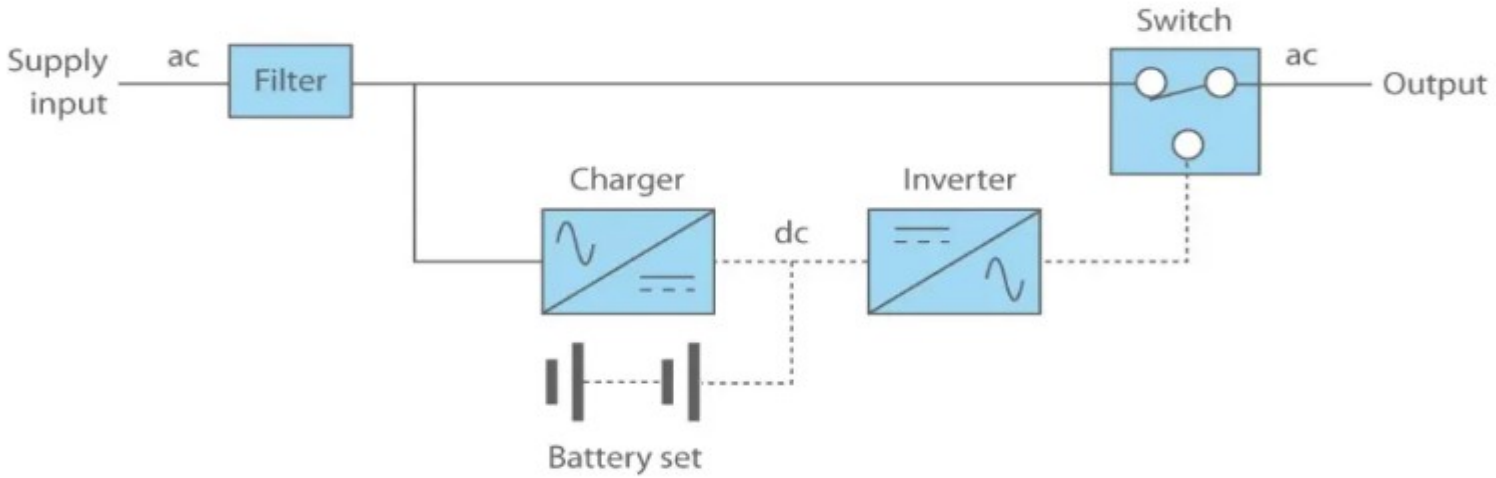
**Charging Process**: When battery is charging, electrons flows from cathode to anode through separator and current flows from anode to cathode.
**Discharging Process**: Electrons flows from anode to cathode through separator and current flows from cathode to anode.



# Block Diagram Components OF UPS :

## 1.Rectifier 2.Battery 3.Inverter 4.Static switch 5.Load





W Wikipedia, the free enc...
Uninterruptible power ...




IQS Directory
Uninterruptible Power Supply | UPS ...




EG Engineers Garage
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ElProCus
Types of Unir



The New York Times



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What are the differences?

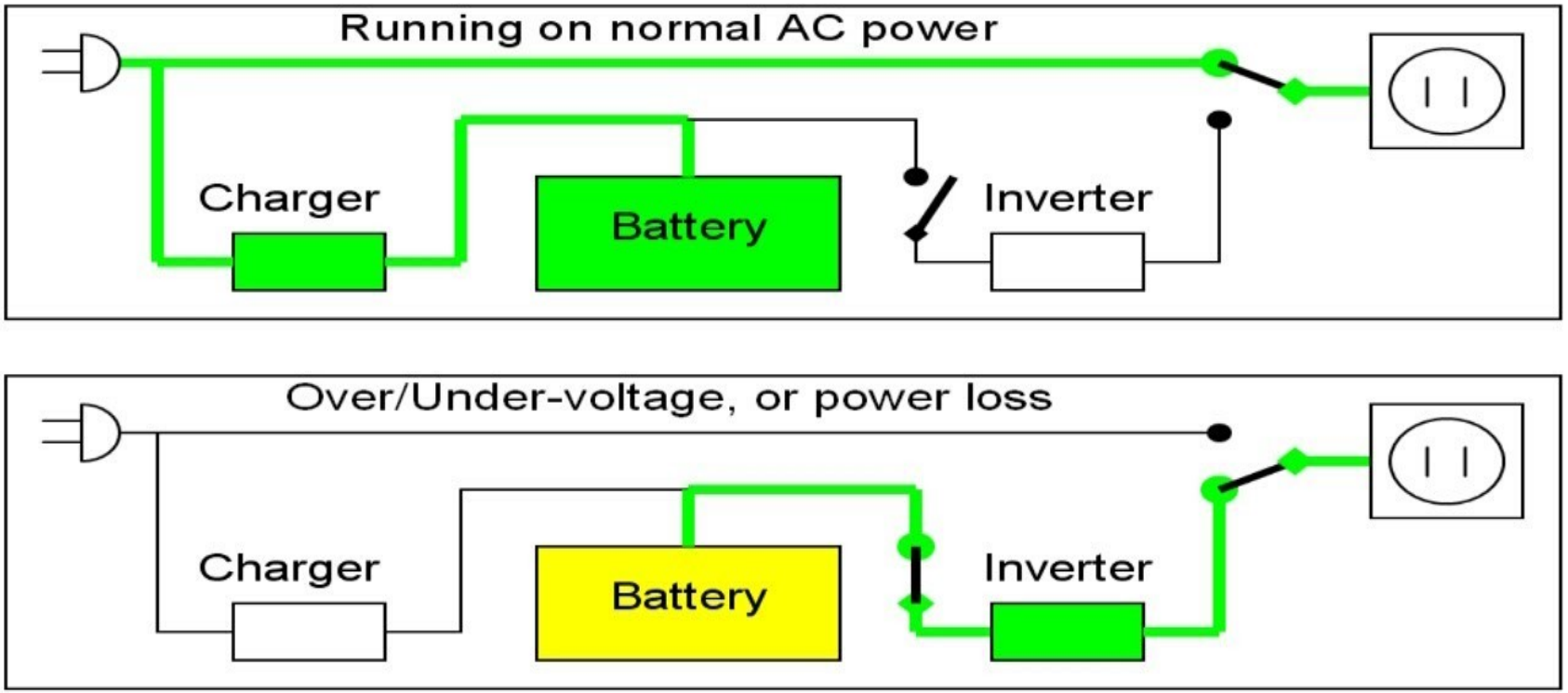


# Uninterrupted Power supply (UPS)

A standby UPS resorts to battery backup power in the event of common power problems such as a power cut. When incoming utility power drops below or surges above safe voltage levels, the UPS switches to DC battery power and then inverts it to AC power to run connected equipment.

Note: Uninterrupted power supply (UPS) is nothing but

UPS = (9V Battery charging application) + (inverter of making 9V DC to low voltage AC )+ (Low voltage AC to 230 V step up Transformer )



# Operation of UPS

## 1. Normal Power Supply Condition

- AC mains supply is available.
- AC supply is converted into DC by the rectifier.
- DC output:
  - Supplies the inverter
  - Charges the battery
- The inverter converts DC back to AC.
- The load receives continuous and regulated AC supply.

## 2. Power Failure Condition

- When AC mains supply fails:
  - Rectifier stops working.
  - Battery supplies DC power to the inverter.
- Inverter converts DC from battery into AC.
- Static switch ensures instantaneous transfer without interruption.
- Load continues to operate without power break.

## 3. Power Restoration Condition

- When AC supply returns:
  - Rectifier resumes operation.
  - Battery gets recharged.
  - Load is again supplied through mains via inverter.

## Applications of UPS

- **Computers and data centers**
- **Medical equipment**
- **Communication systems**
- **Industrial control systems**

# Electrical Safety Standards

- It is essentially important to take precautions when we are working with electricity and using electrical appliances. Here are some of the basic precautions ar mentioned for safe usage of electrical appliances:

- Follow the manufacturer’s instructions
- Replace or repair damaged power cords
- Keep electrical equipment or outlets away from water
- Use insulated tools while working
- Don’t overload your outlets
- Shut – off the power supply
- Avoid extension cords as much as possible
- Avoid the usage of flammable liquids
- Use electric tester
- Display danger board
- Usage of proper ladder
- Usage of circuit breakers or Fuse
- Use ceiling on live wire
- Precaution during soldering